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Review

Battery Lifetime Prognostics

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Lithium-ion batteries have been widely used in many important applications. However, there are still many challenges facing lithium-ion batteries, one of them being degradation. Battery degradation is a complex problem, which involves many electrochemical side reactions in anode, electrolyte, and cathode. Operating conditions affect degradation significantly and therefore the battery lifetime. It is of extreme importance to achieve accurate predictions of the remaining battery lifetime under various operating conditions. This is essential for the battery management system to ensure reliable operation and timely maintenance and is also critical for battery second-life applications. After introducing the degradation mechanisms, this paper provides a timely and comprehensive review of the battery lifetime prognostic technologies with a focus on recent advances in model-based, data-driven, and hybrid approaches. The details, advantages, and limitations of these approaches are presented, analyzed, and compared. Future trends are presented, and key challenges and opportunities are discussed.

Introduction

Limited fossil fuel reserves and climate change challenges provide a strong impetus for developing clean transportation systems, renewable energy sources, and smart grids. Lithium-ion batteries have been widely used in those fields. As one of the most expensive components, they play an important role and should be carefully monitored and operated. Both the electric vehicles and the infrastructure of renewable energy systems and smart grids require long battery lifetime to achieve economic viability. Battery degradation during operation is one of the most urgent and difficult issues, which become the limiting factor in battery lifetime. Due to the complex degradation mechanisms, the battery lifetime varies significantly under different operating conditions.

A lithium-ion battery is a dynamic and time-varying electrochemical system with nonlinear behavior and complicated internal mechanisms. As the number of charge and discharge cycles increases, the performance and life of the lithium-ion battery gradually deteriorate.[1] There are many different causes for battery degradation, including both physical mechanisms (e.g., thermal stress and mechanical stress) and chemical mechanisms (e.g., side reactions).[2] Figure 1 illustrates the most common battery degradation mechanisms. Different degradation mechanisms contribute to the battery degradation, and they can be divided into two main degradation modes[3,4]: (1) loss of lithium inventory, which is caused by consumption of lithium ions through side reactions. (2) Loss of active material, which results in a loss of storage capacity. More specifically, loss of active material is mainly caused by graphite exfoliation, binder decomposition, loss of electrical contact owing to corrosion of current collectors, and electrode particle cracking. Loss of lithium inventory is mainly caused by solid electrolyte interphase (SEI) film formation and decomposition, electrolyte decomposition, and lithium plating. It is worth mentioning that these degradation mechanisms are highly related to materials.[5] For example, the

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The increasing energy demands of a growing population and the challenges of climate change provide a strong driving force for transportation electrification and smart grid development. As one of the most widely used energy storage devices, lithium-ion batteries play an important role in those fields. One of the most urgent issues in lithium-ion batteries is degradation. Automakers have set 15 years in service as the goal for hybrid and electric vehicles. Storage batteries used in renewable energy systems and smart grids also require long lives. A long battery lifetime is critical to achieving the economic viability in electric vehicles, renewable energy, and smart grid infrastructure. However, battery degradation is a complex electrochemical process, which includes many electrochemical side reactions, such as solid electrolyte interphase, electrolyte oxidation, salt decomposition, particle fracture, and active material dissolution. Accurate predictions of the remaining battery lifetime at different operating conditions are essential for the battery management system to avoid potentially dangerous battery failures and guarantee reliable and efficient operation. The remaining battery lifetime information is also critical for battery second-life applications. This paper provides

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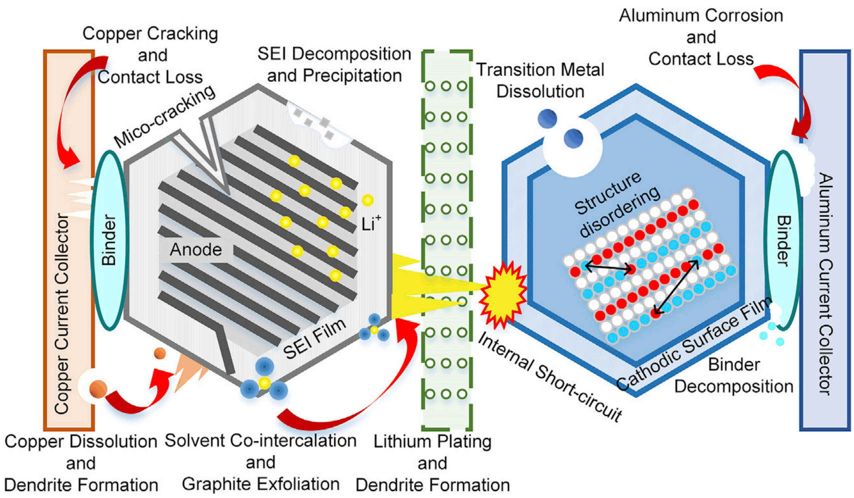

Figure 1. Main Degradation Mechanisms in Lithium-Ion Batteries

Lithium-ion battery degradation mechanisms can be divided into two categories: (1) loss of lithium inventory, such as solid electrolyte interphase growth and lithium plating, and (2) loss of active material, such as graphite exfoliation.
Credit Adapted from Birkl et al.[8]

a comprehensive review of the development of battery remaining useful lifetime (RUL) prognostic techniques. Upcoming challenges and future research directions are identified and discussed.

working voltage of graphite anode is lower than the electrochemical window of commonly used electrolytes, which leads to the formation of SEI film.[6] However, there would be no SEI film formation in the lithium titanium oxide (LTO) anode because LTO's potential is located in the electrochemical window of the electrolyte. Another example is that the volume change of the lithium iron phosphate (LFP) cathode is smaller than that of the lithium manganese oxide (LMO) cathode, and therefore, its structural deformation is also smaller.[7] Besides the material difference, degradation mechanisms vary greatly under different operating conditions and different battery designs. For example, lithium plating is very likely to occur during fast charging, while it rarely occurs during discharge. For battery designs, small cathode particle sizes result in small stresses and therefore less particle cracking but also result in more cathode material dissolution due to high specific surface area.

Due to the complex nature of the battery degradation process, predicting the remaining battery lifetime becomes an extremely challenging task. However, this is essential for the battery management system to ensure reliable operation and timely maintenance and is also critical for battery second-life applications.

When the battery degrades to a certain point, for instance, if a battery can only retain 80% of its initial capacity,[9–11] the battery should be retired to ensure the safety and reliability of the battery-powered systems. As an essential energy storage device in the electrified transportation systems and smart grids, battery failure can cause system malfunction. Certain severe battery failures, such as thermal runaway, can cause an intense energy release that results in fire or explosion. However, due to the uncertainties in environmental and load conditions, it is difficult to predict the battery degradation rate. Moreover, the parameters directly related to the battery aging state are internal variables, which are difficult to measure with sensors. Thus, it is of vital importance to study the degradation behaviors and construct a degradation model to estimate the state of health (SOH) and remaining useful lifetime (RUL). RUL is usually defined as the number of charge and discharge cycles remaining until a failure threshold is reached. An RUL prediction method aims to use historical and

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present performance data to forecast the future state and provide a warning before battery failure occurs. Reliable and accurate RUL prediction is of great significance in ensuring the performance, safety, and economy of the battery-powered systems.

RUL prediction has attracted increasing attention in recent years. Some studies have reviewed existing RUL prediction methods and categorized them according to the models and algorithms used. According to the existing literature, RUL prediction methods include model-based, data-driven, and hybrid approaches. Model-based approaches aim to establish a mathematical model to describe the degradation trajectory based on battery dynamics. However, it is challenging to make a trade-off between model complexity and prediction accuracy. On the contrary, data-driven approaches attempt to extract hidden correlations from a large amount of data and predict the RUL without a battery mathematical model. The data-driven approach does not need to analyze the system mechanism and is feasible and practical when large amount of data is available. In recent years, hybrid approaches that combine the model-based and data-driven approaches have also been proposed because they can combine the relative strengths of different methods.

Recent advances in hardware and software tools, such as the significantly improved computing power of graphics processing units (GPUs) and newly developed software based on them, provide an opportunity to incorporate sophisticated data-driven methods into battery RUL prediction. Hybrid approaches with online updating and error correction capabilities have also been developed. Most of the available reviews on battery RUL prognostics do not include the latest research results of model-based, data-driven, and hybrid approaches.[12–16] In a recent article, data-driven RUL prognostic technologies are reviewed.[17] However, model-based and hybrid approaches are not included in this paper. In another recent study, the options and challenges of the mechanistic model, data-driven model, and hybrid model for lithium-ion battery diagnosis were reviewed.[18] This paper presented an excellent systematic review of battery state of charge (SOC) and SOH estimation, and accurate SOH estimation is very important because it provides useful information for RUL perdition. By including the latest achievements in recent years, we provide a comprehensive review of model-based, data-driven, and hybrid approaches. The state-of-the-art of RUL prognosis technology is thoroughly reviewed, and the research gaps are identified. In particular, the hybrid approach is discussed in detail, and an algorithm flowchart is given to elucidate the details. In this review article, we introduce three publicly available battery datasets and two types of health indexes. We examine the model-based, data-driven, and hybrid approaches that have been used in battery RUL prediction. The algorithm structure of each approach is summarized, and the algorithm features are compared. At the same time, we highlight current research gaps in the literature and suggest the challenges from four perspectives to encourage future research in this area.

Data Acquisition

The performance degradation of a battery is related to various mechanisms, and the declining trajectory is nonlinear. Therefore, battery aging data are required to construct an RUL prediction algorithm and validate its stability and accuracy. In the early days of lithium-ion batteries, the lifetime of the battery was short. Therefore, regular testing could generate sufficient aging data. For instance, the capacity falls to 80% of its original value under 0.5 C–1 C charge-discharge profile within 160 cycles in Saha et al.[19] and 600 cycles in He et al.[20] and Xing et al.[21] With the rapid development of battery technology, battery lifetime has been greatly improved. In a recent work,[22] the capacity decreased only 4% within 1,000 cycles under a 3.6 C charge-discharge profile. The performance degradation in this data is very small, and the data are therefore not suitable as training data. Consequently, accelerated aging tests are needed to obtain battery aging data in a short period of time, with different charge-discharge rates, and ambient temperature being the most commonly selected stress factors. However, conducting battery aging tests is time consuming, and extracting battery features from different aging cycles requires a complex and expensive signal acquisition system. Therefore, many researchers have utilized publicly available battery datasets to verify their RUL prediction algorithms.

Battery Datasets

A widely used battery dataset is available from the Prognostics Center of Excellence at NASA Ames.[19] The NASA battery dataset provides six groups of experimental data. Batteries were charged and discharged at different temperatures, and the impedance was recorded after each cycle. Among them, the most commonly used datasets are B0005-B0007 and B0018. To collect these datasets, batteries were subjected to three different operational profiles (charge, discharge, and electrochemical impedance spectroscopy) at room temperature. After each charge-discharge cycle, an electrochemical impedance spectroscopy (EIS) test was carried out to measure the impedance and obtain internal parameters that reflected the battery aging.

Another widely used battery dataset, CS2, is provided by the Center for Advanced Life Cycle Engineering (CALCE) at the University of Maryland.[20,21] The CS2 battery dataset was obtained using prismatic cells with $LiCoO_2$ cathodes and is often used for RUL prediction. CS2 contains six types of experimental data from batteries that all underwent the same constant current-constant voltage (CCCV) charging process. The discharge currents were set differently for the six datasets. For instance, the discharge current was set to 0.5 C in the type 1 dataset and 1 C in the type 2 dataset.

Recently, Severson et al.[22] provided a large publicly available battery dataset, which consists of 124 LFP-graphite cells. The cells were cycled to 80% of their initial capacities under various fast-charging conditions ranging from 3.6 to 6 C in an environmental chamber (30°C). The cells were then charged from 0% to 80% SOC with one-step or two-step charging profiles (e.g., 6 C charging from 0% to 40% SOC, followed by a 3 C charging to 80% SOC). Then, all cells were charged from 80% to 100% SOC with 1 C CCCV step to 3.6 V and subsequently discharged with 4 C CCCV step to 2.0 V, and the cut-off current was set to C/50. During the cycling test, the cell temperature was recorded and internal resistance was obtained at 80% SOC.

Identification of the Battery Degradation State

In lithium-ion batteries, the conversion between electrical energy and chemical energy is achieved through repeated lithium-ion intercalation and deintercalation. Side chemical reactions can cause irreversible losses of electrolyte, active electrode materials, and lithium inventory; therefore, the capacity decreases over cycles. RUL prediction requires a health indicator (HI) that reflects the capability of a battery to deliver the specified performance compared with a new battery and quantifies the battery's degradation state. Multiple attributes can be used as HIs. In the existing studies, capacity or internal resistance is often used as an HI. By predicting the future trends of these two HIs, the RUL can be calculated as the time interval between the current state and the predefined failure threshold. For instance, the failure threshold is reached when the battery capacity falls to 80% of the initial value, or the internal resistance becomes 1.3 times its initial value.[23] Since these HIs reflect the physical battery degradation directly, they are referred to as direct HIs (DHIs).

There are two major challenges with health monitoring of batteries: (1) the online measurement of battery capacity requires a complete charge-discharge process, which is difficult in many applications, and (2) the predictability of DHI is low, especially for DHIs extracted from early cycle data with negligible performance fade.[22] Therefore, many refined HIs (RHIs) have been proposed. The first type of RHIs, which are extracted from battery voltage, current, and temperature data, can be easily obtained online, such as the mean voltage drop[24] and the time interval of equal discharging voltage difference series.[25] The second type of RHIs, which analyze the available data thoroughly, has better predictability than the DHI, such as the summary statistics (e.g., minimum, mean, and variance) of the change in discharge voltage curves[22] and the sample entropy of the measured voltage sequence.[26]

A degradation modeling process is required after extracting the RHI from the original battery measurements. Degradation modeling has two main purposes: (1) to evaluate the correlation between RHI and DHI (usually battery capacity) and validate the effectiveness of the proposed RHI, and (2) to establish a relationship between RHI and DHI. Therefore, after degradation modeling, the battery RUL can be effectively predicted based on the future RHI trajectory.

Classification of RUL Prognostic Methodologies

In recent years, significant progress has been made in developing RUL prognosis. Several articles have summarized common RUL prediction approaches adopted in battery and machinery prognostics and classified them into various categories.[13,14,27–29] As mentioned previously, RUL prediction methods can be generally divided into model-based, data-driven, and hybrid approaches. However, there are small differences in their classification criteria and naming rules. For instance, filtering approaches are grouped into data-driven approaches in Wu et al.[29] but are grouped into model-based approaches in Sutharssan et al.[14] In addition, there is no comprehensive review of battery RUL prognosis that includes the most recent research advances. Therefore, we have developed an improved classification procedure specifically for battery RUL prediction. The developed procedure comprehensively summarizes the battery RUL prediction methods available in the literature and groups them into three categories, (1) model-based, (2) data-driven, and (3) hybrid approaches. The model-based approach relies on physics-based modeling of the degradation behavior (also referred to as the physics-of-failure model), or it builds an empirical model to describe the declining trajectory of the system. The model is typically a set of algebraic and differential equations or an empirical equation. An important feature of this approach is that the model is designed for a specific system (i.e., the prediction model is different for batteries and for bearings). The data-driven approach uses statistical theories or machine learning techniques to derive a predictive model directly from the measured data instead of building a particular physics-based model. Unlike the model-based approach, the data-driven approach is more applicable to different applications (e.g., the same data-driven algorithm can be applied to both bearing and battery, and only the parameters need to recalibrated). The hybrid approach combines the relative strengths of the above two approaches.

Model-Based Approaches

The model-based approach aims to establish a mathematical model that describes the degradation behavior of a battery. One way of doing this is to build a complex

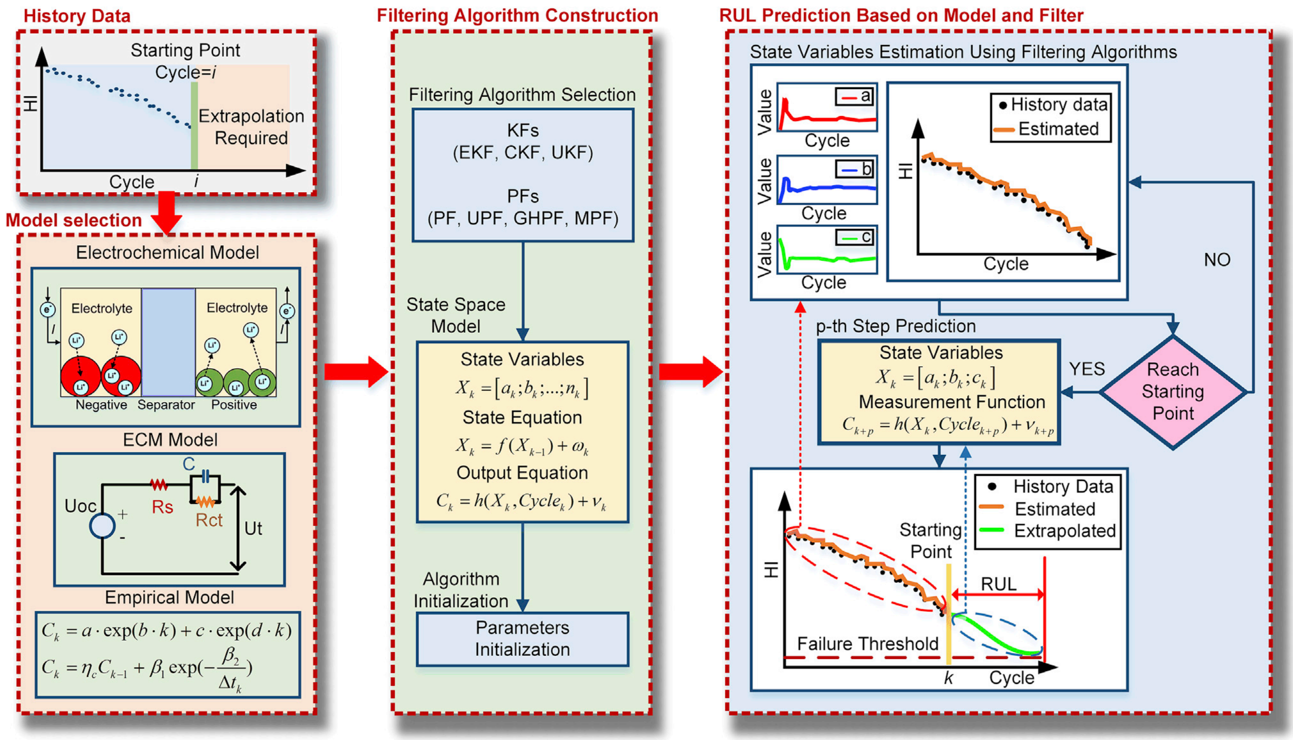

Figure 2. Flowchart of the Model-Based Algorithm

First, history data are obtained, and the last data point is defined as the prediction starting point (see block history data). Based on the data characteristics (e.g., available internal information and degradation trajectory), a suitable model (i.e., electrochemical model, ECM model, or empirical model) is chosen (see block model selection). Then, a filtering algorithm is selected, and the model is transformed into the state-space form. The parameters are also initialized (see block filtering algorithm construction). Finally (see block RUL prediction based on model and filter), based on the filtering algorithm, historical data (black dots) are utilized to estimate the state variables and update the model (solid orange line). For illustration purposes, we assume a simple three-parameter model for RUL prediction, and the model parameters are *a*, *b*, and *c*. Here, assume that the total number of historical data is *k*, then the model parameters are updated until reaching the prediction starting point (yellow column at cycle *k*), and the final state variables are $X_k = [a_k; b_k; c_k]$. The updated model is used for RUL prediction by extrapolating the measurement equation (solid green line).

mechanistic model or an empirical regression model, then to predict future battery performance by extrapolation. However, the battery degradation behavior is nonlinear. For long-term predictions (multiple-step ahead, more than 50 steps), the mechanistic model with side reactions can maintain high accuracy in RUL predictions, but the empirical model with fixed model parameters cannot track the performance change accurately and could lead to large errors.[21] One way to improve long-term prediction is to combine the mathematical model with a filtering algorithm,[30] so that the model parameters can be dynamically updated. As shown in Figure 2, a suitable model is firstly selected based on the available data. Then, the filtering algorithm is constructed and initialized. Finally, the model parameters are updated with filters using historical data, and the updated model is used for RUL prediction. The model-based approach can be divided into four groups: mechanistic models, equivalent circuit models (ECMs), empirical models, or fused models.

Mechanistic Models

Mechanistic models are based on the electrochemical processes of the battery. Reaction kinetics and porous electrode theories are employed to establish the mechanistic model. In an early study,[31] a semi-empirical model was developed for battery capacity fade prediction. However, this model establishes an empirical correlation among model parameters and is limited to a specific battery system (e.g., Sony 18650 cells). Later on, by assuming the capacity loss was caused by the side reaction on the negative electrode surface, the P2D model was combined with a semi-empirical side reaction expression to form a capacity fade model.[32] The proposed model used the Bulter–Volmer kinetic equation to calculate the side reaction rate and the film resistance increment. The developed model can predict battery cycle life, but it can only be used under low charge-discharge current (i.e., lower than 1 C) since the lithium-ion transport in electrolyte was neglected. To solve this problem, a generalized first principle-based model was developed to simulate battery cycle life behavior in Ning et al.[33] Similar to Ramadass et al.[32] only the side reaction at the anode and electrolyte interface was considered, and the anode film growth causes the capacity degradation. In order to use this model at harsh charge-discharge conditions, the transport in both solid-phase and electrolyte phase was incorporated. The developed model was designed for charge/discharge rate up to 2 or 3 C. However, it had only been validated under 1 C condition. In electrochemical models, certain model parameters are dependent on temperature. Therefore, an electrochemical-thermal coupled model was developed to describe battery behavior at varying temperatures.[34] The coupled model can simulate battery behavior under low C rates and dynamic load profiles over a temperature range of 25°C to 40°C, and the changes in battery parameters (e.g., solid-phase diffusion time constants and initial stoichiometric number of the electrodes) at different aging cycles can be used as features for RUL prediction. In a recent study,[35] a particle filter (PF)-based algorithm was combined with an electrochemical model. Battery internal parameters directly related to the SOH, such as electrolyte volume fraction, specific interfacial area, and exchange current density, were selected and treated as state variables and were updated using the latest available data. Compared with traditional empirical models with meaningless fitting coefficients, the updated parameters in this model were battery internal physical quantities. With the updated parameters, better stability and prediction accuracy are achieved. To reduce model complexity, a reformulated pseudo 2D (P2D) model was developed for RUL predictions.[36] By fitting the power-law expression to the estimated effective transport and kinetic parameters from different cycling data, battery RUL was predicted based on the extrapolation of the power-law expression. By using the voltage data from 50–500 cycles, the voltage curve predicted by the model was in good agreement with the actual data at cycle 1,000, which shows a promising future in real-world applications.

The SEI layer formed mainly in the first few cycles acts as a protective layer between the anode and the electrolyte. Studies have shown that the growth and formation of the passive SEI layer, which consumes the cyclable lithium, are one of the main causes of battery capacity fade.[37] However, the mechanism of SEI formation is complex and still not fully understood.[38,39] Although many researchers developed continuum or macroscopic models (P2D model and single-particle model) to describe the SEI formation empirically or semi-empirically, the heterogeneity across the surface of the anode is neglected. To better describe the formation of the SEI layer at the molecular level, Methekar et al.[40] used kinetic Monte Carlo simulation to addressed surface heterogeneity for the first time. The simulation results were in good agreement with the experimental observations, and this study provides a good understanding of the mechanism of SEI film formation. Besides the Monte Carlo simulation, several approaches, such as molecular dynamics[41] and density functional theory,[42] have been proposed to study the SEI formation. Due to the high computational complexity, these methods can only be applied in a short period of time and at a small length scale. To reduce computational cost, Röder et al.[43] proposed an advanced multi-scale modeling method that couples a macroscopic single-particle electrode model with a kinetic Monte Carlo model. The proposed model can be used as a powerful tool not only to explore the mechanism of SEI formation but also to predict and even control SEI formation. In a recent study, Röder et al.[44] systematically investigated three algorithms for directly coupling the macroscopic (i.e., continuum) model and the kinetic Monte Carlo model in electrochemical systems. This study provides useful guidance for building multi-scale simulation techniques that can achieve a balance between computational efficiency and accuracy.

Equivalent Circuit Models

By analyzing the physical and chemical reactions in batteries, the ECM is created based on the combination of circuit elements that produce the same electrical behavior as the battery. Therefore, circuit analysis methods can be used to build mathematical models that describe the battery dynamic response and degradation behavior. The internal resistance gradually increases due to the growth of the SEI layer. EIS data can be used to identify ECM parameters, and a regression model can be used to predict battery RUL. Several groups have developed different internal resistance growth models and employed a PF algorithm for parameter updating.[23,30,45,46] The internal impedance shows a linear relationship with capacity at 1 C current. By taking advantage of this relationship, the internal impedance was used to obtain RUL.[45] Based on the internal resistance growth model, some techniques for improving the performance of the filtering algorithm were proposed. For instance, a Rao–Blackwellized PF framework was used for model updating to reduce the variance in state estimation.[47] In another example, a mutated PF was developed for battery RUL prediction to solve the problem of particle diversity loss, which is also known as sample impoverishment.[48] However, without a feedback mechanism, the mutated PF generates mutated particles blindly. Therefore, an enhanced mutated PF method was proposed, which took into account the prior information about the high-likelihood area and used a dynamic feedback mechanism to explore the posterior space more efficiently.[49] In another example, a regularized auxiliary PF was developed to estimate and update the parameters of an exponential resistance growth model.[50] By regularizing the empirical distribution, the regularized auxiliary PF enhances particle diversity, and a rejection resampling scheme is used to increase the robustness. However, in real-time applications, EIS tests are time consuming, and the equipment may not be available. Therefore, a recent work employed a fractional-order ECM to estimate model parameters using the input current and output terminal voltage of the battery.[51] The parameters obtained under different battery aging stages were used to reconstruct the EIS spectrum, and the reconstructed EIS data were used to establish a regression model that describes the growth in internal resistance.

Empirical Models

An empirical model is established by analyzing the correlations in large amounts of data. This model incorporates common features in battery degradation behavior. It uses different regression models to fit the declining trend or builds an appropriate empirical formula to describe and forecast the degradation state.

Two empirical models were developed for battery RUL prediction,[52] namely, the weighted Ah aging model and event-oriented aging model. Under standard conditions, a certain amount of energy can be put into a battery until it reaches its lifetime threshold. In the weighted Ah aging model, the criterion of battery failure is the point at which the weighted cumulative Ah value exceeds the threshold. In actual applications, there are specific events that cause lifetime loss, for instance, over-charging and over-discharging. The event-oriented aging model evaluates the impact on lifetime caused by each event and adds up all the events to obtain battery RUL. Under low-current conditions, a quadratic polynomial regression model was developed to fit the degradation trend of nickel-metal hydride batteries.[53] By selecting the time, temperature, and SOC during battery testing as model parameters, an empirical internal resistance model was established that can be used for calendar and cycle-life prediction over a temperature range of 40°C–70°C.[54,55] Similarly, different HIs were utilized to construct empirical models—for instance, the fully discharged voltage and internal resistance,[56] the rate of temperature change,[57] and the battery discharge curve.[58]

The empirical model is simple because it ignores the internal mechanism of batteries. Based on historical data, the empirical model establishes the mapping between RUL and battery characteristics (e.g., capacity or internal resistance). Such correlations vary with time mainly due to battery aging and changes in working conditions, such as temperature and current rate. Therefore, the empirical model is often combined with a filtering algorithm to update the model parameters based on the latest available data. Usually, exponential and polynomial functions are used to build empirical aging models with the number of parameters ranging from two[59] to four.[60] To choose the appropriate formula for the empirical model, the curve shape of the chosen empirical model should be similar to the historical data. In the case of limited historical data, empirical models with fewer parameters should be used to ensure the accuracy of parameter identification. In the case of sufficient historical data, empirical models with more parameters can be used. In the existing studies, Kalman and PFs are the most commonly used methods for parameter estimation. Kalman filters (KFs) are recursive filters that contain a prediction step and an updating step. They are suitable for time-varying linear systems.[61] However, the empirical model is usually nonlinear. Therefore, extended KF[62] and unscented KF[63] are adopted. Another commonly used filter is the PFs, which are proposed under the framework of recursive Bayesian estimation.[64] PFs can handle most nonlinear and non-Gaussian noise problems and are therefore suitable for battery RUL prediction. Various empirical models have been used as the state-space equation for the filtering algorithm, such as the exponential model,[59,63,65] the logarithmic model,[66] the polynomial model,[67] the ensemble model,[21] and the Verhulst model that has fewer parameters than traditional empirical models while maintaining accuracy.[68] Based on the above algorithm framework, further studies were conducted to improve prediction accuracy. Different discharge rates, which affect battery capacity and its degradation rate, were incorporated into a traditional exponential model to develop a new empirical model.[69] PFs with abnormal detection[70] and constrained KF[71] were used to reduce the noise in the original data. An exemplar-based conditional PF was developed,[72] which combines data from multiple batteries to calculate particle weights. Singular value decomposition was employed to generate sigma particles in the unscented transformation to solve the ill-condition problem in unscented PF.[73]

There are two main problems in the standard PF algorithm: (1) in the process of particle propagation, a few particles have large weights, while others have negligible weights (i.e., sample degeneracy), and (2) particles with small weights are abandoned in the resampling process (i.e., sample impoverishment). To address these issues, many studies have enhanced the standard PF algorithm to enrich particle diversity and improve prediction performance. The traditional resampling method was replaced with the artificial fish swarm algorithm, which optimizes particles based on preying and swarming behaviors.[74] A heuristic Kalman algorithm, which combines a

Table 1. Evaluation of Model-Based Methods

Method	Advantages	Disadvantages
Mechanistic	• Considers the internal electrochemical process • High accuracy • High generalization ability	• Requires expert knowledge • Difficult to establish or identify model parameters • High computational costs
ECM	• Considers the aging mechanism to some extent (i.e., describes the internal resistance growth trend)	• Requires expensive equipment • EIS test accelerates the aging process
Empirical	• Easy to establish • Wide applicability	• Model parameters need to be updated • Low generalization ability
Fused	• Increases useful information • Performs well in the early stages	• Hard to set the fused failure criterion • Increases error sources

Gaussian generator with a Kalman updating framework to optimize particles before resampling, was used.[60] The spherical cubature KF and the Gauss–Hermite KF were used to provide the importance probability density function for PFs.[75–77] In a novel scheme,[78] the particle-learning framework was utilized to resample the particles before propagating them. By taking current measurements into account, the proposed method avoids particle decay and reduce RUL prediction error. By incorporating the latest measurements, the above methods drive prior particle distribution to the high-likelihood region, which improves particle distribution and the algorithm accuracy.

Fused Models

The aforementioned models can be included in one of the three categories, i.e., mechanistic models, ECM models, or empirical models. To extract more features and information from the available data, fused models that incorporate different types of battery models are developed. The first type of fused model combines different models to build a new model. For instance, a recent example combined the capacity degradation model (i.e., the empirical model) with the internal resistance growth model (i.e., the ECM model). A PF algorithm was used to update model parameters dynamically for RUL prediction. Compared with a single degradation model, the fused model has higher accuracy and can reduce prediction uncertainty at an early stage.[79] The second type of fused model uses different models to calculate model parameters. For instance, a recent study proposed a three-step battery RUL estimation.[80] In the first two steps, the core temperature, SOC, and capacity were estimated based on a thermal model and an ECM model using *sliding mode observers*. In the third step, the above estimated values were used to identify the parameters of an empirical capacity degradation model, and the fitted empirical model was then used for battery RUL prediction.

Comparison of Model-Based Approaches

An in-depth understanding of the battery internal reaction mechanisms is required to establish the mechanistic model. Although these models can capture the battery electrochemical behavior, they have the highest complexity. ECMs and empirical models are easier to build, but they can only estimate the short-term states due to the changes in internal parameters during cycling. Therefore, a filtering method is often used to update the model dynamically and improve prediction accuracy. By combining different types of these models, fused models can be developed to improve model performance. The advantages and disadvantages of model-based approaches are shown in Table 1.

Data-Driven Approaches

Data-driven approaches use historical monitoring data directly to predict the degradation trend of a battery while its mechanisms and propagation rules remain unknown. Instead of using a specific physics-based model, this method establishes a mathematical model or obtains weight parameters based on only training data. Since this method avoids building a complex physics-based model, it is more flexible and applicable and has attracted extensive attention from researchers worldwide. Battery RUL prognosis based on data-driven approaches can be divided into three categories according to whether they are based on artificial intelligence, statistical analysis, or signal processing.

Algorithm Structures

In the existing literature, four algorithm structures have been used in the data-driven approaches (Figure 3).

Iterative Structure. The iterative structure uses multiple historical data to predict the value of the next step, and long-term predictions are achieved iteratively, as shown in Figure 3A. For a capacity time series $\mathbf{c} = \{c_1, c_2, \ldots, c_N\}$, phase space reconstruction is employed to transform the original $N \times 1$ vector into an $L \times m$ dimensional matrix $\mathbf{c_p}$ as follows[84]:

$$\mathbf{c_p} = \begin{bmatrix} c_1 & c_{1+\tau} & \cdots & c_{1+(m-1)\tau} \\ c_2 & c_{2+\tau} & \cdots & c_{2+(m-1)\tau} \\ \vdots & \vdots & & \vdots \\ c_L & c_{L+\tau} & \cdots & c_{L+(m-1)\tau} \end{bmatrix}$$

where m is the embedding dimension and τ is the time delay. L is the rows of matrix $\mathbf{c_p}$ and $L = N-(m-1)\tau$.

The reconstructed matrix $\mathbf{c_p}$ and the original vector $\mathbf{c}$ are used as input data and corresponding target data, respectively, to form the training data set D:

$$\mathbf{D} = [\text{input}|\text{target}] = \left[\begin{array}{cccc|c} c_1 & c_{1+\tau} & \cdots & c_{1+(m-1)\tau} & c_{2+(m-1)\tau} \\ c_2 & c_{2+\tau} & \cdots & c_{2+(m-1)\tau} & c_{3+(m-1)\tau} \\ \vdots & \vdots & & \vdots & \vdots \\ c_{L-1} & c_{L+\tau-1} & \cdots & c_{N-1} & c_N \end{array}\right]$$

Then, $\mathbf{D}$ is used to train various data-driven models to obtain the prediction function $h(.)$. After training, the iterative structure is expressed as follows:

$$\hat{c}_{N+1} = h(c_L, c_{L+\tau}, \ldots, c_N)$$

where $\hat{c}_{N+1}$ is the predicted value of the next step.

The predicted value $\hat{c}_{N+2}$ can be obtained by substituting $\hat{c}_{N+1}$ into the input vector and is given as:

$$\hat{c}_{N+2} = h(c_{L+1}, c_{L+\tau+1}, \ldots, \hat{c}_{N+1})$$

Thus, long-term prediction can be carried out iteratively.

Non-iterative Structure. For a historical dataset of battery capacity, $\mathbf{c} = \{i, c_i\}_{i=1}^{N}$, where i is the cycle number and c_i is the corresponding battery capacity, the training dataset D of the non-iterative structure is formed as:

$$\mathbf{D} = [\text{input}|\text{target}] = \left[\begin{array}{c|c} 1 & c_1 \\ 2 & c_2 \\ \vdots & \vdots \\ N & c_N \end{array}\right]$$

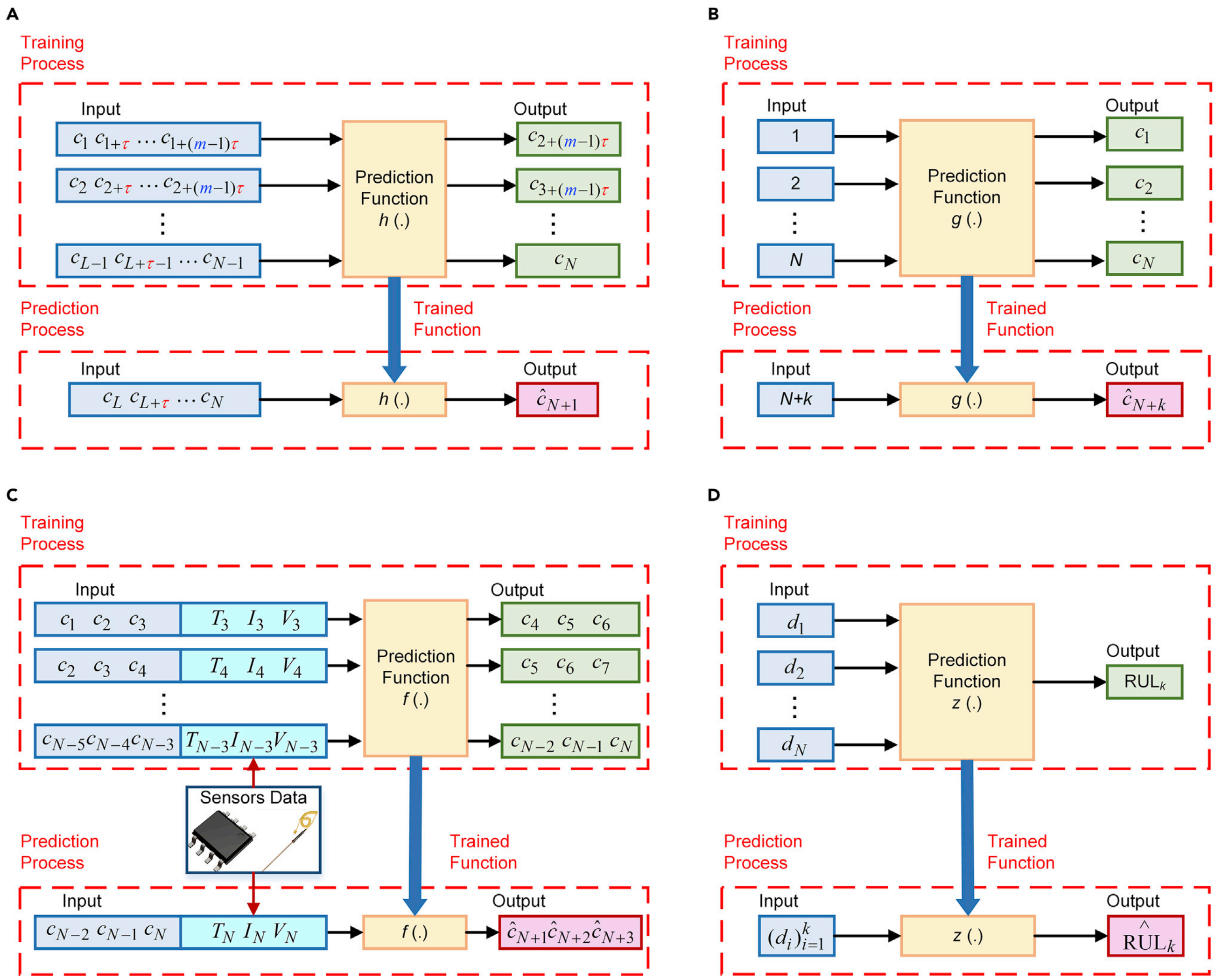

Figure 3. Algorithm Structures of the Data-Driven Approaches

(A) Diagram of the iterative structure. For a capacity time series $\mathbf{c} = \{c_1, c_2, \ldots, c_N\}$, phase space reconstruction is employed to transform the $N \times 1$ vector into an $L \times m$ matrix $\mathbf{c_p}$. Here, m is the embedding dimension, and τ is the time delay. $L = N - (m - 1)\tau$ is the rows of matrix $\mathbf{c_p}$. Each training example is composed of a $1 \times m$ dimensional feature $\{c_k, c_{k+\tau}, \ldots, c_{k+(m-1)\tau}\}$ and a corresponding output $c_{k+1+(m-1)\tau}$. The predicted value $\hat{c}_{N+1}$ can be obtained by substituting feature $\{c_L, c_{L+\tau}, \ldots, c_N\}$ into the trained function $h(.)$. Thus, multiple-step-ahead prediction can be carried out iteratively.

(B) Diagram of the non-iterative structure. The training dataset is formed as $\{i, c_i\}_{i=1}^{N}$, where i is the cycle number, and c_i is the corresponding battery capacity. Thus, the forecasting value at cycle $N + k$ can be directly obtained by substituting the cycle number into the prediction function $g(.)$.

(C) Diagram of the short-term structure. The training dataset of this structure is composed of several measured signals with historical capacities. Here, it is assumed that three sensory data (temperature, current, and voltage) at cycle N (T_N, I_N, and V_N) are required, and that three historical capacities (c_{N-2}, c_{N-1}, and c_N) are needed to predict the next three steps (c_{N+1}, c_{N+2}, and c_{N+3}).

(D) Diagram of the direct structure. By establishing a relationship between historical data $\mathbf{d} = \{d_1, d_2, \ldots, d_N\}$ and corresponding RUL_N, this structure obtains battery RUL directly. In the existing literature, the training data can be historical capacity,[81] the combination of historical capacity fade and power fade,[82] or the battery terminal voltages of t cycle.[83]

The prediction function $g(.)$ is obtained by training different data-driven models, and the non-iterative structure is constructed as follows:

$$\hat{c}_{N+k} = g(N + k)$$

Thus, as shown in Figure 3B, the forecasting value at cycle $N + k$ can be directly obtained by substituting the cycle number into the prediction function $g(.)$.

Short-Term Structure. In both iterative and non-iterative structures, online measured data are not required. In contrast, the short-term structure incorporates several online signals (e.g., temperature, current, and voltage) with historical data to form the training dataset D, as shown in Figure 3C. For a battery capacity dataset c={c1,c2,…,cN}, the temperature, current, and voltage at cycle *k* are denoted as T_k, V_k, and I_k. Assuming that three historical data points are needed to predict the next three steps, the training dataset D can be constructed as:

$$\mathrm{D} = \left[\begin{array}{cccccc|ccc} c_1 & c_2 & c_3 & T_3 & I_3 & V_3 & c_4 & c_5 & c_6 \\ c_2 & c_3 & c_4 & T_4 & I_4 & V_4 & c_5 & c_6 & c_7 \\ \vdots & \vdots & & & \vdots & & & \vdots & \\ c_{N-5} & c_{N-4} & c_{N-3} & T_{N-3} & I_{N-3} & V_{N-3} & c_{N-2} & c_{N-1} & c_N \end{array}\right]$$

Consequently, with the trained prediction function *f*(.), the short-term structure is constructed as follows:

$$(\hat{c}_{N+1}, \hat{c}_{N+2}, \hat{c}_{N+3}) = f(c_{N-2}, c_{N-1}, c_N, T_N, I_N, V_N)$$

By considering the online measurements in each cycle, the short-term structure can exploit more useful information and has high accuracy. However, as can be seen from the above equations, this structure can only predict a few steps ahead; thus, it is not suitable for long-term RUL prediction and is rarely used in the literature.

Direct Structure. Unlike forecasting the degradation trajectory from historical data (e.g., capacity), the direct structure directly outputs the battery RUL (e.g., remaining cycle number), as shown in Figure 3D. This is achieved by establishing a relationship between historical data d and RUL, for instance, the Wiener process obtains the probability density function (PDF) of RUL directly. The direct structure can be expressed as:

$$RUL_k = z(d_i)_{i=1}^{k}$$

Algorithm Structure Classification. All four algorithm structures can be found in various artificial intelligence approaches, while the statistical approaches mainly use three types of structures: direct, iterative, and non-iterative. Each statistical algorithm has a definite form, and its structure can be easily identified. Generally, the autoregressive integrated moving average (ARIMA), which is often used in time series analysis, is an iterative structure, the gray model (GM), which is effective under insufficient data, is a non-iterative structure, and the Wiener process (WP), which describes a continuous-time stochastic process, is a direct structure.

Artificial Intelligence

Artificial intelligence learns from samples to establish complex relationships between factors that are difficult to represent in mathematical models. This method is easy to implement, and the models can be built quickly due to the recent advances in computer hardware and software tools.

For battery RUL prognosis, the training data for artificial intelligence mainly consist of cycling numbers, current, voltage, temperature, and capacity. Because the expected output values are known, it can be classified as supervised learning. Under the framework of supervised learning, a HI that represents battery performance is first selected. Then, the artificial intelligence system establishes mapping relations between the HI and battery RUL using its powerful nonlinear fitting ability.

Naive Bayes Model. The Naive Bayes (NB) model is the simplest form of a Bayesian network. An NB model is basically a classifier that uses the Bayes' theorem as the discriminant function $f_c(x)$.[85] Therefore, the new observation is assigned to the class whose discriminant function $f_c(x)$ obtains the maximum value.

Selina[81] used battery data from NASA, which consists of different load and temperature conditions, to train an NB model for RUL prediction. The RUL values of different batteries are the output variable of the training data and are divided into several classes, whereas the corresponding historical capacity data of each battery is the input variable of the training data. To evaluate the robustness of the algorithm, external conditions such as the load profiles and ambient temperature are not used in the NB model training. When a historical capacity data sequence is obtained, the NB model calculates the discriminant function value for each class, and assigns the obtained data to the class with the maximum value. However, the NB model was trained under the constant current condition. Therefore, the prediction accuracy could deteriorate when the operating conditions are changed.

Support Vector Regression. In traditional regression models, the loss function is defined as the error between f (x) and its real value *y*. On the contrary, support vector regression (SVR) skips errors that are smaller than ϵ and the loss is counted only when $|f(x) - y| > \epsilon$. By constructing the Lagrangian loss function, the solution of SVR can be obtained due to its convexity.[86] If the data have a nonlinear trend, the mapping function is employed to transform the low-dimensional input space into a high-dimensional feature space. Thus, the nonlinear regression problem in the input space becomes linear in the feature space.

By considering the correlations among current, voltage, and operation time, energy efficiency and battery working temperature were proposed as two RHIs. These RHIs were determined by a dimensionality-reduction technique that reduces the computational load while preserving all physical battery characteristics. Based on these RHIs, an RUL prediction model was built using SVR.[87] Based on real-world driving profiles, the dwell-time counting and the rain flow counting methods were adopted to build the training data. The training data were later used to train the SVR model for RUL prediction.[88] By taking into account battery degradation under different temperatures, a PF algorithm was used to dynamically update the parameters of the SVR model based on the online data.[89] When new measurements are added to the training data, the SVR model is retrained. However, the retraining of the SVR model is very time consuming. Therefore, instead of retraining, an online SVR model that is updated by an incremental learning algorithm was proposed,[90] and the model update can be done quickly. In another example,[91] the time interval of the equal charge and equal discharge voltage difference were extracted as two RHIs and were used to construct the training set for SVR. Then, a novel data-driven approach that combined feature vector selection with SVR was proposed. In the feature vector selection method, a subset of feature vectors is selected to represent the original dataset. In this way, the redundant training data are eliminated, and the search scope of the SVR model is narrowed.

Machine learning approaches can estimate battery RUL based on various measurements, such as current, voltage, and temperature, and, therefore, can be used at various operating conditions.[13] However, as the training data increase, the computational complexity also increases, which is evident in the SVR algorithms.[92] Within the support vector framework, Patil et al.[93] proposed a two-stage RUL prediction that combines classification and regression models. At the early stage, the classification model was used to obtain a rough RUL value, and when the battery approached its end-of-life threshold, the regression model was used to estimate an accurate RUL value. By taking an additional classification step, the time-consuming regression model was only activated near the end of battery life. Therefore, the developed method greatly reduces the computational cost, thereby speeding up the computation.

Relevance Vector Machine. Based on the sparse Bayesian learning theory, Tipping[94] developed the relevance vector machine (RVM) in a form similar to SVR. Compared with SVR, RVM has a higher degree of sparsity and can provide probabilistic predictions.

In one study,[24] the mean voltage drop was extracted and used as an RHI. Box-Cox transformation was employed to increase the linearity between RHI and capacity. RVM was used to establish an RUL prediction model based on the proposed RHI and the transformed capacity data. In another case, in order to improve the long-term prediction capability, an improved RVM model that can be incrementally updated using online data was proposed. The improved RVM does not use all existing data but only uses relevant vectors and new online data to generate new training data. The computational efficiency is almost twice that of the standard retraining model.[95] In another work, Zhao et al.[96] extracted the last hidden layer of a deep belief network to generate the training data for the RVM model. This method effectively extracts data features, reduces computational burdens, and achieves good prediction performance.

Gaussian Process Regression. As a stochastic process, Gaussian processes (GPs) contain finite sets of variables and each set of variables has a joint Gaussian distribution. Therefore, a GP is fully defined by its covariance function and mean function.[97] The priori regression analysis of GP data is defined as Gaussian process regression (GPR) or Kriging.[98]

In an earlier study,[30] GPR was used to predict the future trajectory of battery internal resistance, and RUL was derived from the linear correlation between capacity and internal resistance. The standard GPR method consists of a zero mean function and a diagonal covariance squared exponential function. Results showed that although GPR can fit a nonlinear degradation trend and achieve good prediction performance (relative error = 9.3%) for a 16-step-ahead prediction, its long-term prediction accuracy is low. Another study had a similar conclusion that as the number of prediction steps increases, the prediction accuracy of GPR decreases rapidly.[99] To improve the long-term performance of GPR, new covariance functions and mean functions were incorporated into GPR, such as a linear mean function[100] and a combined covariance function that consists of the periodic covariance function and the diagonal squared exponential covariance function.[101] A recent study proposed two methods to exploit prior information and improve the performance of GPR.[102] First, an explicit mean function based on a known battery degradation model was used to exploit prior knowledge of capacity fading. Second, multi-output GPR was used to analyze the correlations between different cells and exploit available data effectively. By considering self-discharging and capacity regeneration behavior, the capacity fade trajectories of batteries have the property of multimodality and can be divided into multiple segments. By using different GPRs to fit individual trajectory segments, the multimodality was handled well.[103]

Artificial Neural Networks. The artificial neural network (ANN) method is inspired by the biological neural network, such as the brain.[104] Generally, an ANN consists of an input layer, one or more hidden layers, and an output layer, and its basic component is the artificial neuron. For example, for a layer of a neural network with *n* neurons, the activation of each of these *n* neurons depends on the activations of the neurons in the previous layer. The outputs from the neurons in the previous layers are multiplied by corresponding weights and added together. The sum is used as the input to the neuron in the current layer. A bias is added to this sum, followed by the application of a nonlinear activation function to generate the output for this neuron. After constructing the ANN, an algorithm is employed to determine the weights of the interconnections based on training data.

Multilayer feedforward neural networks (FNN) trained by the back-propagation (BP) learning algorithm are the most widely used type of neural network.[105] The BP-FNN training procedure consists of signal forward-propagation and error back-propagation steps that adjust the weights. Through an iterative process, the weights of connections between neurons are updated to minimize the loss function. This supervised training process continues until the error reaches an acceptable value.[106] In an earlier study,[107] a BP-FNN with an output layer of two neurons representing the discharge and charge capacity, and a BP-FNN with an input layer of five neurons and an output layer of one neuron to establish a relationship between historical and forecast capacity data were constructed.[108] Results showed that the long-term performance of the BP-FNN was poor. At different aging stages, batteries showed different charge and discharge voltage profiles. Therefore, importance sampling was used to select representative voltage data from the voltage curve during charging, which improved computational efficiency. The selected voltage data were then used by an FNN for RUL prediction.[83] Besides BP-FNN, other ANN structures were also used in battery RUL prediction. For example, as shown in Figure 3C, the input to the neural network has been rearranged to include the latest online temperature, current, and SOC, as well as the historical capacity data. With this input, a recurrent neural network is trained, and the mean square error of predicted capacity is less than 0.5 Ah.[109] However, the recurrent neural network model can only perform short-term capacity predictions (3 steps ahead). Therefore, it is unable to achieve long-term RUL prognosis. In one study, to improve the prediction accuracy, an adaptive recurrent neural network was developed, which updated the weights adaptively using the recursive Levenberg–Marquardt method.[110] In another example, the Long Short-Term Memory neural network was used for battery RUL prediction. The resilient mean square BP technique designed for mini-batch training was used to train the model. A dropout technique was employed to prevent overfitting.[111] In another study,[25] the time interval of equal discharging voltage difference series was extracted from online monitoring data as an RHI. Using the proposed RHI, an ensemble learning strategy was incorporated with a monotonic echo state network to improve prediction accuracy and stability. Based on the algorithm framework developed by Liu et al.,[25] a recent study quantified the uncertainty in RUL prediction by assuming that the RUL values followed a Weibull distribution.[112]

Neuro-Fuzzy Models. Neuro-fuzzy models (NFs) are an integration of fuzzy logic and neural networks that combines the semantic transparency of fuzzy *if-then* rules with the learning, adaptation, and generalization capabilities of neural networks.[113,114] Adaptive-network-based fuzzy inference systems are one of commonly used NF models for modeling nonlinear functions or predicting chaotic time series. In one application of NF model, three different predictors were constructed for battery RUL prognosis based on different rule selection strategies, namely, subtractive clustering, fuzzy-c-means, and grid partitioning.[115] By using battery test data from NASA, various data-driven methods were compared in terms of root mean square error and prediction error. The NF model with subtractive clustering outperformed the other methods (i.e., neural networks, group method of data handling, and random forests) in terms of the root mean squared error values in RUL prediction.

Statistical Approach

Statistical approaches build statistical models based on empirical knowledge and available data to predict the trend of battery capacity degradation. In a probabilistic framework, this approach constructs a random coefficient model or a stochastic process model to describe battery capacity decline based on historical measurements. Because it does not rely on expert knowledge, the statistical modeling approach is easy to implement. Moreover, this approach incorporates random variances to describe the uncertainties in battery degradation behavior caused by momentary fluctuations, cell-to-cell variations, and measurement errors. Therefore, the statistical approach can describe the uncertainty in battery degradation effectively and provide accurate RUL prediction results.

Autoregressive Integrated Moving Average. A stationary linear stochastic process can be represented by a small number of autoregressive (AR) and moving average (MA) terms. For nonstationary random process, Box et al.[116] developed an ARIMA model that employs an initial differencing step (corresponding to the "integrated" part of the model) to remove the nonstationary component. The ARIMA model can be approximated by a high-order AR model. Since the computational efficiency of AR is higher than ARIMA, AR model is often employed for battery RUL prediction.[46,117] By setting the order of moving average processes to zero and removing the differencing step, the ARIMA model is reduced to an AR model. The model order has a significant impact on the performance of the AR model. Therefore, the particle swarm optimization algorithm was utilized to determine the optimal order and improve the prediction accuracy.[118] The battery capacity degradation rate accelerates as the battery ages, and the linear AR model cannot predict this nonlinear trend. To solve this problem, a nonlinear degradation (ND) factor was proposed and was combined with the standard AR model (ND-AR).[119] However, when sample size and prediction starting points change, the generalizability of ND-AR becomes poor. Therefore, the ratio of the current cycle number to the entire life cycle number was proposed as a modified ND factor in another study. The proposed method showed better RUL prediction performance than the ND-AR model on both NASA and CALCE battery datasets.[120]

Grey Model. The grey system theory was proposed by Ju-long Deng in 1982 and is capable of solving small-sample and poor-information problems.[121,122] The grey model, GM(1,1), is one of the most widely used grey prediction models. For a non-negative historical time series, smoothing algorithms (e.g., accumulated generating operation) are used to reduce the randomness of the raw data. Then, the smoothed data are assumed to satisfy a first-order ordinary differential equation. By solving the unknown parameters of the ordinary differential equation, the discrete-time form of the time series is obtained. The mechanism of battery degradation involves many factors and remains poorly understood, but the degradation trend caused by these factors can be quantified using HIs. The whole process can be considered as a ''grey cause and white results'' system, which can be modeled using grey system theory.[122] The battery capacity degradation is nonlinear. Therefore, the standard GM(1,1) based on a first-order ordinary differential equation is not accurate in long-term predictions. To improve the prediction accuracy, in one study, the least-squares method with forgetting factors was employed to estimate and update the model parameters, and greater weights were given to the newly measured data.[123] In another study, a GM(1,1) and a residual GM(1,1) were used to describe battery degradation trend and prediction errors, respectively.[124,125] Moreover, the parameters of the above model were dynamically updated using online measurements to achieve high prediction accuracy.[125]

Wiener Process. A degradation process $\{X(t), t > 0\}$ can be modeled as a WP with a linear drift.[126] The lifetime of the system is often defined as the first hitting time when the degradation exceeds a certain threshold.[127] Based on the definition of the first hitting time, the RUL L_k of a system at time t_k is defined as the remaining time from t_k to the first time when $X(t)$ exceeds the failure threshold w. For a Wiener process, the PDF of the first hitting time follows an inverse Gaussian distribution.[128] Therefore, L_k also follows an inverse Gaussian distribution and its PDF can be obtained. For example, a WP model with nonlinear drift and scaling factors was built to predict battery RUL. The state-space form of a WP containing a drift parameter was constructed, and RUL is predicted by two steps (offline and online). During the offline step, the model parameters were estimated using the maximum likelihood estimation method based on historical data, and the online step employed a PF algorithm to update the state variable (i.e., drift parameter) of the WP model. Finally, battery RUL was obtained based on the assumption that the first hitting time of the WP follows an inverse Gaussian distribution.[129] Following the above two steps (offline and online), a WP model with three state variables (i.e., random drift, a diffusion coefficient, and measurement errors), was established to achieve better accuracy.[130] In another study, a WP model with two state variables (i.e., the amplitude and slope of an exponential degradation model) was developed.[131] These state variables were first identified using offline measurements and then updated using online data. A recent study extracted the duration of the CV mode and the duration of the CC mode from the CCCV charging data as two RHIs. The battery degradation process was modeled by a two-dimensional WP to obtain the PDF of RUL.[132]

Entropy Analysis. The sample entropy is a measure of the complexity of time series signals.[133] A large value of sample entropy represents a high complexity of the information, while a small value indicates more regularities in the original data. Sample entropy is a modification of approximate entropy, which not only overcomes the self-matching problem in approximate entropy but is also less dependent on the data length. Based on the discharge curves of batteries at different cycle numbers, multi-scale entropy was employed to extract the sample entropy of five timescales and analyze the predictability of battery RUL. As the cycle number and timescale increase, the multi-scale entropy also increases, which indicates a decrease in RUL predictability.[134] Therefore, using one model to predict battery RUL will lose accuracy as cycle number increases, and a model-switching algorithm is desirable for long-term prediction. Entropy analysis is more commonly used as a feature extraction method in the existing literature. The entropy values calculated from battery measurements, such as voltage, current, and temperature, are used as RHIs, which have been shown to have a stronger correlation with degradation performance.[26,135] These RHIs are later used as training inputs for various data-driven algorithms to build RUL prediction models.

Signal Processing

Various signal-processing methods have been used to extract useful information from the raw data for diagnosis and prognosis purposes.[136] Signal processing methods have three main purposes: (1) filtering out the distortion, (2) removing redundant data that is irrelevant for the system diagnosis or prognosis, and (3) transformation of the signal to decouple or whiten its features. Unprocessed battery monitoring signals contain momentary fluctuations and measurement noises. Signal processing methods are great tools for battery data processing, and RUL prediction can be achieved based on the processed data.

Discrete Wavelet Transform. The discrete wavelet transform (DWT) method decomposes a signal into several components with different frequencies and provides a powerful tool for analyzing nonstationary signals. The DWT method is derived based on the continuous wavelet transform through the discretization of the mother wavelet, and it can be easily implemented in digital computers.[137,138] A recent work utilized a dynamic stress test schedule to produce nonstationary battery capacity signals and decomposed the raw signals into approximate and detailed signals based on the DWT method.[139] The detailed signals fluctuated much higher in aged cells than in new ones. Therefore, the coefficients of the detailed signals were used to establish an empirical model for RUL prediction. However, this method requires a dynamic performance test to calculate the RUL, which is time consuming. In addition, the proposed empirical model is linear and will have large errors in long-term predictions. Although very few studies have used only DWT for RUL prediction, this method is widely used to separate signals of different scales or eliminate noise in battery sensor signal preprocessing.

Comparison of Data-Driven Approaches

The data-driven approaches include artificial intelligence, statistical analysis, and signal processing methods, each with their own characteristics. Table 2 gives a summary of the DHIs, RHIs, and algorithm structures used in specific studies, as well as the advantages and disadvantages of each approach.

Hybrid Approaches

A key issue in the data-driven methods is that insufficient or biased training data can lead to inaccurate predictions or completely false results. Meanwhile, model-based approaches require expert knowledge to develop physical models and are less flexible than data-driven methods. However, model-based approaches require fewer data and, due to their insensitivity to external uncertainty, are stable and robust. In recent years, hybrid approaches combing model-based methods with data-driven methods to obtain accurate predictions have become a research hotspot in battery RUL prognosis, attempting to take advantage of the relative strengths of both approaches. In general, hybrid approaches can be divided into three categories according to the purposes: (1) improving the performance of filtering methods, (2) generating future observation data, and (3) processing raw data.

Improving the Performance of Filtering Methods

As the battery degradation continues, the model parameters need to be updated to maintain prediction accuracy. This is commonly done by filtering algorithms. However, there are several problems related to initialization and sample impoverishment in filtering algorithms. The initialization problem usually exists in model parameter estimation methods, such as least squares, KFs, and PFs. Traditionally, the initial values are set empirically or even randomly. However, inappropriate initial values can slow down the convergence of the algorithm and could also lead to divergence. The Dempster–Shafer theory was utilized to initialize the PF parameters,[20] which reduces convergence time and increases prediction accuracy at the early stage of battery lifespan.

Table 2. Evaluation of Data-Driven Methods

Method		Health Index	Algorithm Structure	Advantages	Disadvantages
Artificial Intelligence	NB	• DHI (capacity)[81]	• Direct[81]	• Concise • Insensitive to missing data	• Based on attribute independence assumption
	SVM	• DHI (capacity, internal resistance)[89,90,82] • RHI (energy efficiency, temperature)[87]	• Direct[82] • Short-term[87,89] • Iterative[88]	• Capable of dealing with local minimum, nonlinear, and small sample size problems	• Kernel functions need to satisfy Mercer criterion • Lack of sparseness
	RVM	• DHI (capacity)[95,140,141] • RHI (the mean voltage drop)[24]	• Iterative[95] • Non-iterative[24] • Curve fitting[140,141]	• Better sparsity • Free from Mercer restraint • Avoids overfitting and underfitting	• High computational load with large datasets • Not appropriate for long-term prediction • Lack of stability
	GPR	• DHI (capacity)[100–103]	• Non-iterative[100–102] • Iterative[103]	• Handling high-dimension and small sample size datasets	• High computational load with large datasets • Lack of sparseness
	ANN	• DHI (capacity, internal resistance)[107,108,109–112,142] • RHI (discharging curve)[25,112] • RHI (terminal voltage curve)[83]	• Iterative[25,108,110,142] • Non-iterative[107,142] • Short-term[109] • Direct[83]	• Strong learning and nonlinear processing ability • Integrates information	• Requires sufficient training data • Complex structure • High computational complexity
	NF	• DHI (capacity)[115]	• Iterative[115]	• Can solve the inherent inaccuracy and fuzzy characteristics • Uses ANN to optimize the inference structures	• Requires sufficient training data • High computation complexity
Statistical Analysis	AR	• DHI (capacity)[46,117–120]	• Iterative[46,117–120]	• Utilizes a small amount of historical data • Good real-time performance • Easy to identify model parameters	• Not appropriate for nonlinear signals • Lack of robustness and long-term accuracy
	GM	• DHI (capacity)[123,124] • RHI (discharging curve)[125]	• Non-iterative[124] • Iterative[123,125]	• Capable of dealing with small simple sizes, poor information, and uncertainty problems	• Not appropriate for non-smooth data • Background value has a large impact on prediction accuracy
	WP	• DHI (capacity)[129,131,143] • RHI (charging curve)[132]	• Direct[129,131,132,143]	• The distribution of the first hitting time can be formulated analytically • Suitable for non-monotonic process modeling	• Based on Markov property • Not appropriate for nonlinear and time heterogeneous data
	Entropy	• RHI (sample entropy)[135]	–	• Extracts features of data effectively • Simple for onboard application	• Sensitive to the quantity and quality of data
Signal Processing	DWT	• RHI (wavelet decomposition of the voltage curve)[139]	• Direct[139]	• Suitable for nonstationary data • Preserves signals characteristic while suppressing noise	• Wavelet function parameters are hard to choose

Standard PF uses resampling techniques to avoid particle degradation by eliminating particles with small weights and duplicating those with large weights. However, this will produce a lot of large-weight particles, whereas the number of small-weight particles is almost zero, which causes sample impoverishment. To avoid this problem, the SVR algorithm was adopted in resampling steps to maintain the diversity of particles,[144,145] as shown in Figure 4A. The basic idea is to rebuild the posterior distributions and obtain re-weighted particles based on SVR, where the training data are the particles and their corresponding weights. Another recent

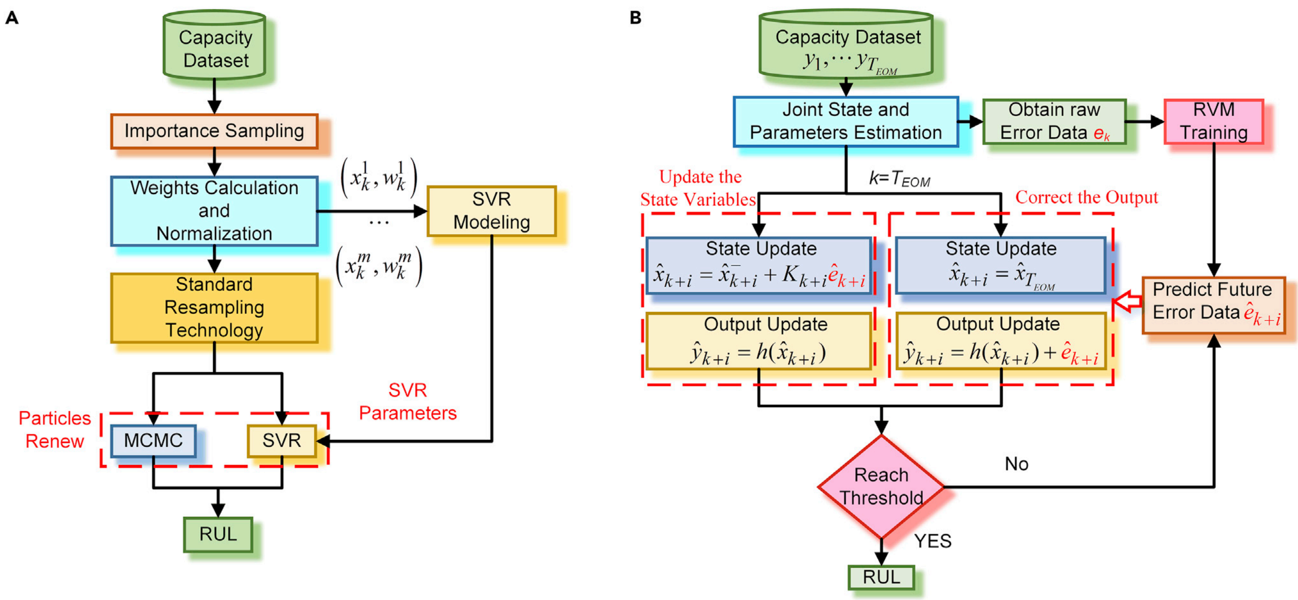

Figure 4. Hybrid Approaches for Battery RUL Prediction

(A) Hybrid approach using a data-driven method to renew particles. One way is to use particle x_k^i and its corresponding weight w_k^i to train the SVR algorithm, and then use SVR to renew particles after resampling. Another way, the MCMC method, is to construct a Markov process to renew particles.
(B) Hybrid approach using a data-driven method to generate a future observation sequence. Here, the historical error e_k is used to train the RVM algorithm, and then the trained RVM is used to predict future error values $\hat{e}_{k+i}$. One way is to use $\hat{e}_{k+i}$ to update the priori state variable $\hat{x}_{k+i}^-$ and obtain the posteriori state variable $\hat{x}_{k+i}$. Thus, the output value $\hat{y}_{k+i}$ is updated using measurement function. In another way, the state variable remains unchanged. Instead, $\hat{e}_{k+i}$ is used to modify the output $\hat{y}_{k+i}$ directly.

study used a Markov chain Monte Carlo (MCMC) method after the standard resampling process to solve particle degradation.[146]

Generating the Future Observation Sequence

During long-term prediction, the state variables (i.e., model parameters) of the empirical degradation model need to be updated, and this is usually done by using a filtering algorithm. However, the future measurements (e.g., battery capacity in the future) are not available, which are required for the updating function of the filtering algorithm for long-term RUL prediction. Therefore, without future measurements, the filtering algorithm can only predict the future trend by simply iterating the state equation based on the last known information and the model parameters remain the same during prediction. The missing future observation data contain the degradation information, and the lack of this data will result in RUL prediction errors. In the literature, two different methods are used to solve the above problem. One way of doing this is to use the data-driven algorithms (e.g., ND-AR[147] and RVM[148]) as the output equation to provide a future observation data series. With the future observation data series, the filtering algorithm can update the state variables (i.e., model parameters) and predict battery RUL. A recent study extended the above work,[147] and the optimized prediction results after filtering were used to update the ND-AR training data, which further improved prediction accuracy.[149,150] Another way of doing this is to use the data-driven algorithms (e.g., RVM) to provide the future residual sequence, as shown in Figure 4B. For instance, Chang et al.[151] used the prediction error series to correct the unscented PF output results, and Zheng and Fang[152] utilized it to update the unscented KF state variables. The purpose of the above methods is to use prediction errors to correct output results or update model parameters. Thereby, prediction accuracy is improved.

Processing Raw Data

Sudden, momentary, and occasional capacity regeneration exists in lithium-ion battery aging process.[153] Such phenomena are correlated with the electrochemical properties of batteries and affected by temperature and load conditions. Degradation trends are affected by capacity regeneration. Therefore, capacity regeneration affects the performance of RUL prediction.[154] It is reasonable to treat raw capacity data as a mixed signal. Various data-driven methods have been adopted to decompose original signals by separating information on different scales to extract useful information. For example, empirical mode decomposition[155,156] and the wavelet decomposition method[157] were utilized to separate raw signals into different parts (e.g., global degradation and local regeneration), and then data-driven methods (e.g., ARIMA and GPR) were adopted to fit these trends, respectively. Finally, the battery RUL was obtained by adding up the individual predicted values. The above combination of the data-driven method for data processing and the data-driven approach for RUL prediction is essentially a data-driven approach, not a hybrid approach. Although the above data processing methods are mainly used in conjunction with the data-driven RUL algorithms, they may also be used to detect and separate different degradation modes in raw data for model-based RUL algorithms. By separating degradation modes, different mechanistic models can be applied for each degradation mode, and the RUL value can be obtained by aggregating prediction values from each model, which reduces the difficulties of building mechanistic models. This combination of the data-driven method for data processing and the model-based approach for RUL prediction is a hybrid approach.

Besides, the acquired battery capacity data often contains various types of noises and measurement errors. Using such raw data to build a capacity fade model or to train a data-driven model can reduce prediction accuracy and even lead to errors. Therefore, it is essential to preprocess raw data to remove noises. Several studies have proposed an improved grey model[158] or discrete grey model[159] to smooth the raw data and reduce the associated randomness. Other studies have sought to remove noises from the raw data. For instance, wavelet denoizing[160,161] and variational mode decomposition[162] were utilized to process data, which effectively removed nonlinear and nonstationary noises. Next, the processed data can be used as the inputs to the model-based approaches for battery RUL prediction. In some studies, RVM was used to obtain the relevant vectors (i.e., data with less noise and measurement errors) from raw historical capacity data. These representative vectors contained battery capacities and corresponding cycling numbers and were used to calculate the parameters of empirical capacity degradation models by the nonlinear fitting. Extrapolation of the established empirical model was employed for RUL prediction.[140,141]

Prognostic Techniques for Calendar Aging and Cycle Aging

For real-world applications, battery capacity gradually decreases during both operation mode (i.e., cycling aging) and storage mode (i.e., calendar aging). In the previous sections, three different approaches (i.e., model-based, data-driven and hybrid) for battery RUL prediction are discussed in detail.

Most of the studies in the previous sections focus on the battery capacity fade under working conditions, and different model-based, data-driven, and hybrid approaches were proposed for RUL prediction. Although the capacity fade, including both calendar aging and cycle aging, is modeled, the above studies excluded long-time rest conditions from experiments in order to accelerate the battery degradation. Therefore, the results of these studies are mainly valid for cycle life predictions.

Calendar aging is defined as the degradation of a battery under idle or storage conditions. Calendar life prediction is very important in real-world applications, because, for example, the battery pack of an electric vehicle spends 90% of its lifetime in storage condition.[163] There are many studies on battery calendar life modeling, and model-based approaches are the most popular methods in calendar aging modeling. Three variables, namely, storage time t, temperature T, and SOC, are commonly used to build the semi-empirical calendar aging model. A general formula of this model is given as follows:

$$Q_{loss,cal}(T, SOC, t) = k_{T,Q_{loss}}(T) \cdot k_{SOC,Q_{loss}}(SOC) \cdot f(t)$$

The effect of storage time $f(t)$ is often modeled using a power function of time (t^z), where the coefficient z varies from 0.5 to 1.[164–166] The temperature dependence $k_{T,\mathrm{Qloss}}(T)$ of calendar aging is often modeled using the Arrhenius equation, and a recent study used the Eyring equation to improve prediction accuracy and reduce parameters.[164] The SOC dependence $k_{SOC,Qloss}$(SOC) is often fitted by exponential functions[163] or polynomial functions.[167] A recent study used a reformulated Tafel equation to build SOC dependence, which introduced the electrochemical dynamics into the proposed calendar life model.[168] Besides, by assuming that the calendar and cycling aging follow the same degradation mechanism under storage and operation, some studies have built calendar and cycle aging empirical models separately, and then combined them to obtain RUL values by superposition.[166,168–170] However, the interaction of these two aging modes was neglected, and these empirical models were only validated in the early stage of battery aging (e.g., the battery capacity only falls to 90%–80% of the initial value).

Without intercalation and deintercalation of lithium-ion, calendar aging can be accurately predicted using model-based approaches. For instance, the root mean square error (RMSE) of predicted calendar life can less than 1% under storage conditions with varying temperature.[163] However, it is worth noting that the calendar aging behavior is not fully considered in the studies from Model-Based Approaches to Hybrid Approaches due to the lack of long-time rest periods in experiments. In other words, calendar aging and cycle aging are considered as two independent behaviors in existing studies, which neglects the interaction of these two different aging modes. Therefore, in future work, data-driven approaches, such as GPR with probabilistic prediction capability, can be used to couple cycle aging and calendar aging instead of simple superposition.

Challenges and Future Trends

In recent years, significant progress has been made in the battery RUL prognosis. Nevertheless, the current research is still in its early stages and focuses on the RUL prediction under specific conditions. Four major challenges remain: battery datasets, first principle-based prognosis, early prediction algorithms, and engineering applications.

Prognosis Using Real-World Battery Datasets

Currently, researchers working on RUL prognosis mainly use battery cycling data from NASA, CALCE, or their own experiments. Most of these battery data were obtained under the CCCV cycling condition, and the batteries were tested at a fixed temperature. However, in most battery-powered applications, working conditions vary greatly, and the batteries experience multiple degradation modes that switch dynamically. Also, with the rapid development of battery technology, battery performance and lifetime have greatly improved. Therefore, battery aging behavior under dynamic and demanding working conditions is much needed. It is worth mentioning that a recent dataset provided by the research team from MIT and Stanford provides the largest publicly available dataset on 124 LFP/graphite cells.[22] These cells were cycled under 72 different fast-charging conditions, and therefore, this dataset is valuable for studying battery degradation under dynamic working conditions. It is highly recommended that researchers use this dataset to develop and validate their methods in future studies.

First Principle-Based RUL Prognosis

For battery RUL prediction, a first principle-based model that captures degradation behaviors with small computational complexity would have good prediction performance under different working conditions. However, it is very difficult to build such a model because there are many coupled nonlinear electrochemical processes in the battery system.[171] The first principle-based models that capture the detailed degradation process are very useful for RUL prognosis. Traditionally, battery degradation is modeled using a homogeneous P2D model based on porous electrode theory.[172] To reduce the complexity of the P2D model, simplified models (e.g., single-particle model) and reformulated P2D models (e.g., based on spectral method) are proposed and combined with semi-empirical degradation models, such as SEI layer growth model,[173] to predict battery degradation. However, the above homogeneous model only provides overall capacity fade and ignores the degradation caused by the inherent heterogeneity of the battery electrodes or caused by some smaller-scale processes. Therefore, it is necessary to model the battery degradation on a smaller length scale and incorporate electrode heterogeneity.[18] In order to do so, mesoscale models, such as kinetic Monte Carlo simulation,[40] have been proposed. It should be noted that this type of model is usually very computationally costly. Therefore, developing multi-scale simulation techniques that combine a macroscopic homogeneous model (e.g., P2D) with a molecular heterogeneity model[44] or use statistical methods in heterogeneity analysis[174] is of great importance to reduce the computational burden. In addition, the battery pack is composed of modules in actual applications. Considerable temperature gradient exists in large-scale batteries, which affects the battery performance. Short circuits caused by mechanical abuses are also a critical problem.[175] Therefore, multi-physics models that combine electrical, thermal, and mechanical behaviors are needed. Although some thermal-electrochemical models[176] and mechanical-electrical models[177] have been proposed, their computational costs need to be significantly reduced for onboard applications.

The first principle-based multi-scale and multi-physics models can greatly improve RUL prediction accuracy. However, given the limited onboard computational resources of the battery management system, effective model reduction techniques must be developed. Therefore, incorporating first-principle knowledge into RUL prognosis provides high performance while reducing the computational cost to a reasonable level for onboard applications is the major challenge. Two promising approaches can be used in future studies. First, mathematical techniques, such as Galerkin and Spectral methods, can be used to reformulate the model, which can effectively reduce the computational cost. In addition, global sensitivity analysis, such as polynomial chaos expansion,[178] can be used to quantify the individual and joint effects of model parameters on the output,[179] and low-sensitivity parameters can be simplified or reduced. Second, data-driven methods can be used as a powerful tool for solving highly nonlinear differential-algebraic equations and partial derivative equations in first principle-based models, which can greatly reduce computational cost. For example, the recurrent neural network can effectively approximate the ordinary differential equation.[180]

Early Prediction Algorithms

An RUL algorithm that can provide early prediction with limited available data is of great importance to prevent battery failure. However, implementing early prediction is a challenge for most available RUL methods. For a given model, up to 40%–70% of the entire lifecycle data are generally required, either to estimate model parameters or to train data-driven methods.[20,21,35,78,181] For lithium-ion batteries, some degradation modes do not manifest in the capacity fade during early cycling. Therefore, the early cycle data have a weak correlation with the RUL. And also, some features might have changed significantly while the capacity remains unchanged.[182,183] For instance, the active material loss of the negative electrode at a delithiated state shifts the discharge voltage curves, but the capacity could remain the same.[22] Three promising methods can be used to achieve RUL early prediction. The first method is to train the model offline using data obtained under similar working conditions and use a small amount of online data (e.g., 20%–25% of the entire lifecycle data[111]) to retrain the model before prediction. The second method is to explore original data thoroughly and develop new RHIs, which have a better correlation with the capacity degradation in early cycles (e.g., a new RHI was developed based on the first five cycles and can be used for prediction[22]). The third method is to develop an algorithm that provides predictions with different accuracies. For example, in the case of limited data, the algorithm uses a classification model to predict RUL with lower accuracy in early cycles, and when the data are sufficient, the algorithm uses a regression model to provide accurate RUL predictions.[22,93]

Engineering Applications

The majority of the existing studies on RUL prognosis were conducted under laboratory conditions, and further study is required under the actual operating conditions. Figure 5 highlights four directions for future research in this field.

Battery Pack RUL Prognosis. In many battery-powered applications, large-scale lithium-ion battery packs are needed, which consist of a lot of cells connected in series, parallel, or series and parallel. Battery cells are electrochemical systems with diversity and complexity. Inconsistencies in the battery pack are caused by differences in materials between cells (e.g., inconsistent particle size distributions of active materials) and different manufacturing parameters (e.g., compacting pressure and drying time).[186] In addition, due to the temperature gradient in the battery pack, cells usually age unevenly.[184] All these factors increase the variations in cell performance. The variations among cells make it difficult to model the degradation for a battery pack accurately. Some studies on battery pack SOH[187,188] estimations have been conducted and have shown that the battery pack model requires more computational resources than a cell model. Moreover, most studies on battery pack SOH estimation are based on empirical methods or ECMs (Figure 5A), and the computational cost is still acceptable. If first principle-based models are used for battery pack RUL prognosis, the computational cost will significantly increase. In addition, with intercell inconsistency in the battery pack, it is challenging to build degradation models and predictive algorithms for battery packs.

Vehicular Cloud Computing Technology. "Cloud computing is a model for enabling convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction."[189] By adopting cloud computing technology, real-time vehicle battery data can be uploaded to a cloud computing center for processing. The powerful computing capability of cloud computing makes it possible to use complicated RUL algorithms that cannot be run

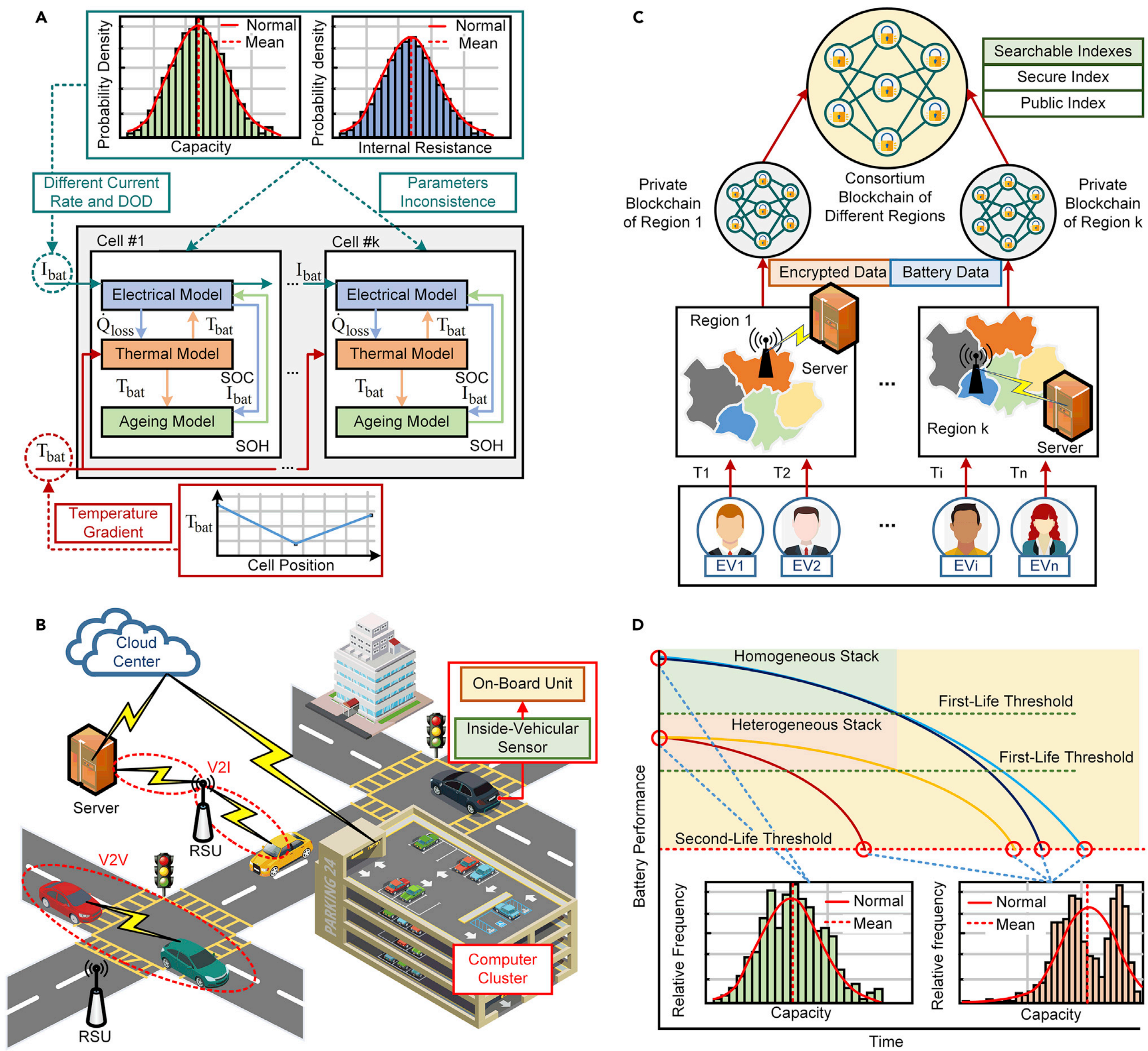

Figure 5. Hybrid Approaches for Battery RUL Prediction

(A) Battery pack RUL prognosis. The top panel shows that the initial capacity and internal resistance of different cells follow a normal distribution. Consequently, even under the same operating current, the current rate and depth of discharge are different for each cell, and for a parallel-connected battery pack, the difference in internal resistance also causes a difference in current values. Moreover, as shown in the bottom panel, temperature gradient commonly exists in the cooling system (e.g., up to 15 K in AC-cooling systems[184]), which leads to a difference in ambient temperature for each cell.

(B) Vehicular cloud computing technology (VCC). The structure of VCC consists of three parts: vehicle, communication, and cloud. In the first part, the travel history is recorded by the inside-vehicular sensor, and the onboard unit, which possesses a broadband wireless communication system, connects the vehicle to the second part. Vehicle-to-vehicle and vehicle-to-infrastructure commutation models are available in VCC. Then, the road-side unit transfers the gathered data to a local server, which finally sends the data to the cloud. An important feature of VCC is that it uses idle resources in participant vehicles to form a data center to store a massive amount of data or to construct a local computer cluster (the parking lot in this image) to collaborate with a cloud computation center.

(C) Blockchain technology. Assume that people from several regions agree to form a league to share their battery data. In order to guarantee data safety and protect driver's private information, two kinds of blockchains are adopted: private blockchains, which store the original full information of each EV, and consortium blockchains, which only store the searchable indexes. Each EV gets a token (T) after registration. When an EV enters region k, the server generates a corresponding vehicle record, which contains private information (i.e., driver's identity [ID] and travel routes) and public information (i.e., battery data) for it. Then, the private information is encrypted and generates a secure index, while the public information generates a public index

Figure 5. *Continued*

directly. The above information is then sent to the private blockchain of region k, and their searchable indexes are sent to the consortium blockchain. Thus, any EV in this league can utilize the battery data of other EVs to train the RUL prediction algorithm. For practical use, the trained algorithm needs to be recalibrated according to the EV's own historical driving data. This is achieved by sending the token T to the consortium blockchain to decrypt the secure index and finding all the battery data that belongs to this EV.

(D) Battery second-life use. The homogeneous stack represents batteries with similar first-life behavior (blue and black lines). However, in second-life use, they show different capacity decreasing trends under identical working conditions. The heterogeneous stack contains batteries with different aging histories during first life, wherein the yellow line cell experienced a mild condition, and the red line cell experienced a demanding condition. It can be seen that differences in the first-life condition greatly affects the second-life performance. Moreover, the bottom panels show that cell-to-cell variation (e.g., capacity) increases during battery aging.[185]

on vehicle battery management systems. In daily use, many vehicles spend hours in a parking lot. These parked vehicles are suitable as nodes in a cloud computing network, and thus, vehicular cloud computing (VCC) technology can be used to leverage resources available in participating cars[190] (Figure 5B). However, the high mobility and wide range of vehicles make VCC more complicated than cloud computing[191] and vehicle traveling data are private, so the sharing of information may leak the travel route information, thereby compromising their privacy. Therefore, an efficient and safe VCC system is needed for vehicle battery RUL prognosis.

Blockchain Technology. A blockchain is a public ledger system whose data are shared among a network of peers.[192] Blockchain technology, which is fully distributed and decentralized, originated as a tool for the Bitcoin cryptocurrency and is well known for its data integrity and security. Blockchain technology can perform data tracking and storage for a large number of devices, and its data are open to everyone. In addition, blockchain can also perform transactions anonymously and protect user privacy. These features make the blockchain technology an ideal choice for information transmission. For example, in electric vehicle (EV) applications, a battery capacity degradation database maintained and accessed by all EV drivers could be established. Therefore, battery data from different driving conditions and areas can be shared among vehicles to obtain more accurate RUL prognosis models (Figure 5C). Currently, there are some research gaps in blockchain technology,[192,193] such as energy efficiency, latency, throughput, and size, and additional research is needed to improve its usability. It is important to build a battery RUL prediction ecosystem based on feasible blockchain technology for engineering applications.

Battery Second-Life Use. Partially degraded batteries removed from the battery-powered systems still have good energy storage capacity and can be used in less demanding second-life applications. Promoting battery second-life use can reduce battery lifecycle costs,[11] and various second-life applications have been proposed.[194] Many studies have shown that second-life batteries have larger cell-to-cell variations than new ones.[185,186,195,196] For instance, the battery cells show similar capacity fade trends in the early cycles, but later, the fading rate of some cells is significantly higher.[197] In addition, the aging trajectory will change its path after reaching a certain point during battery cycling, which suggests that the dominant aging mechanism has changed, and such a point is known as the aging *knee point*.[198,199] After reaching the knee point, the battery cannot be used for long-time second-life applications for safety reasons, because its aging becomes very fast under even mild demanding application.[200] Many experimental results have shown that knee points can appear before or after the battery first life[201–203] (i.e., 80% of the original capacity), and identifying the knee point is challenging. Some studies used empirical methods based on the slope change rate of the capacity to find the knee point.[203,204] In future studies, more electrochemical dynamics should be considered to investigate the changes in the aging mechanism.

The first-life working condition has a significant impact on the second-life use,[199,200] when the degradation mechanisms have changed due to the harsh first-life working conditions. For example, demanding first-life conditions will greatly shorten the second-life of the battery (Figure 5D). Due to the above reasons, the RUL prognosis of the second-life batteries faces many challenges. The first challenge is to obtain the historical record of the battery's first-life working conditions and evaluate their impact on second-life use. The second challenge is to retrain the RUL prediction model for second-life applications using limited data. The third challenge is to identify the aging knee point that could cause sudden failure.

Future RUL Prognostic Techniques and Implementation

There are two major limitations in current RUL prognosis methods. First, for researchers with no expertise in battery electrochemistry, empirical models or data-driven models are natural choices. These models can be built more easily without involving reaction mechanisms, and they are typically built for a specific battery type, such as $LiFePO_4$ (LFP) or $LiNiCoMnO_2$(NCM), under specific working conditions (e.g., CCCV cycling). Therefore, they need to be recalibrated or even rebuilt when battery types or working conditions are changed. Second, for the researchers with backgrounds in materials and chemistry, mechanistic models are more appealing. Mechanistic models capture the electrochemical, thermal, and mechanical behaviors of the battery and are able to predict the battery degradation accurately under different working conditions. Although the first principle-based models provide a deep understanding of the battery mechanisms, it is difficult to build due to the high nonlinearity nature of the battery system. The computational complexity of the first principle-based models also hinders their actual applications.

For future battery applications, capturing the electrochemical, thermal, and mechanical behaviors of the battery is very useful for the battery management system. Therefore, we suggest that the incorporation of first principle knowledge into RUL prognosis is critical for real-world applications, but the computational cost is the greatest concern. Therefore, we conclude that the first principle-based RUL prognosis with low computational complexity is the most promising approach. Two suggestions are provided for future research to reduce model complexity and ensure real-world applications. The first suggestion is to combine the first principle model with the data-driven algorithm to reduce model complexity (First Principle-Based RUL Prognosis). For instance, a recent study proposed a data-driven approach based on the long short-term memory neural network to predict anode potential for lithium plating prevention.[205] Traditionally, anode potential is only available in the first principle-based model with high computational complexity. The proposed long short-term memory neural network model can provide accurate predictions with a much lower computational cost. This demonstrates the potential of data-driven approaches in model simplification. The second suggestion is to integrate intelligent transportation systems to improve computational speed. For instance, cloud computing technology can be used to solve the first principle-based model (Vehicular Cloud Computing Technology).

Here are some guidelines for practical implementations. With adequate computing resources (e.g., during the maintenance of an EV), mechanistic models can be used to thoroughly inspect the internal state of the battery and accurately predict RUL. During battery operations, empirical models and data-driven models are a better choice since they have low computational complexity and can be implemented in the battery management systems. Figure 6 provides useful guidance for selecting empirical models and data-driven models at different stages of the battery life cycle.

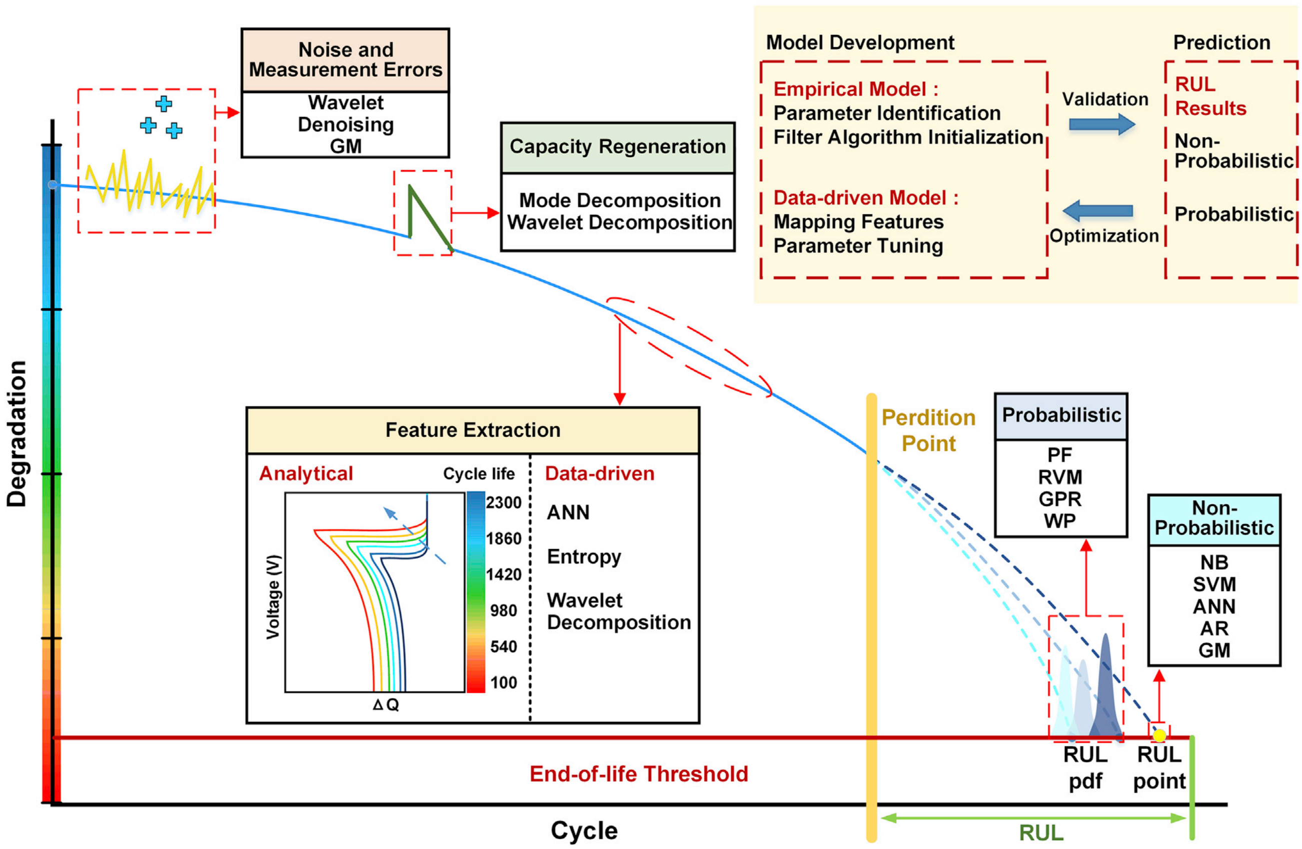

Figure 6. Selection of Empirical and Data-Driven RUL Prognostic Methods at Different Stages

The yellow region (in the upper right corner) shows the general steps for implementing each type of model, which consists of only two steps. In the first step, the parameters of the empirical model are estimated using historical data by nonlinear fitting, and the filter algorithm is initialized (e.g., determining initial states). If data-driven methods are adopted, the correlation between HIs (e.g., capacity) and RUL are mapped, and the parameters are tuned to ensure the generalization capability. In the second step, the established models (i.e., empirical models or data-driven models) are used to predict battery RUL, and their performance is used to optimize the model in turn. Usually, the raw historical data contain noise and sensor errors. Therefore, wavelet denoising and GM can be used. Also, capacity regeneration can occur after a short rest period, and mode or wavelet decomposition can be used to separate these values. The processed data are then used for feature extraction, which increases the correlation between input HIs and output RUL. Two types of methods, analytical (e.g., calculate the change in discharge voltage curves) and data-driven methods (e.g., using ANN and entropy to extract features), are commonly used for feature extraction. The prediction algorithms can be divided into two categories, probabilistic (e.g., PF and GPR) or non-probabilistic (e.g., SVM and AR). The probabilistic methods provide RUL PDFs and confidence bounds for the predicted RUL values, while the non-probabilistic methods only provide a single RUL value. For long-term prediction, the probabilistic methods are a better choice due to the uncertainties from the measurements, the working conditions, and the model itself.

Before model training, measurement noises, sensor errors, and capacity regenerations should be removed or separated by some methods, such as wavelet decomposition. In order to provide the confidence level of the RUL predictions, probabilistic methods, such as PF and GPR, can be used. It should be noted that when the working conditions change (e.g., from first-life use to second-life use), the empirical model and the data-driven model must be retrained or even rebuilt.

In order to compare different prognostic methods with regards to various battery performance metrics, Figure 7 divides different RUL prediction techniques into two groups: (1) ECMs that capture the internal impedance and mechanistic models. Both of them rely on expert knowledge, and (2) empirical and data-driven models that do not need expert knowledge. An understanding of battery internal impedance behavior is needed to build the ECM,[18] as shown in Figure 7A. And the understanding of electrochemical dynamics is needed for the mechanistic models. ECMs

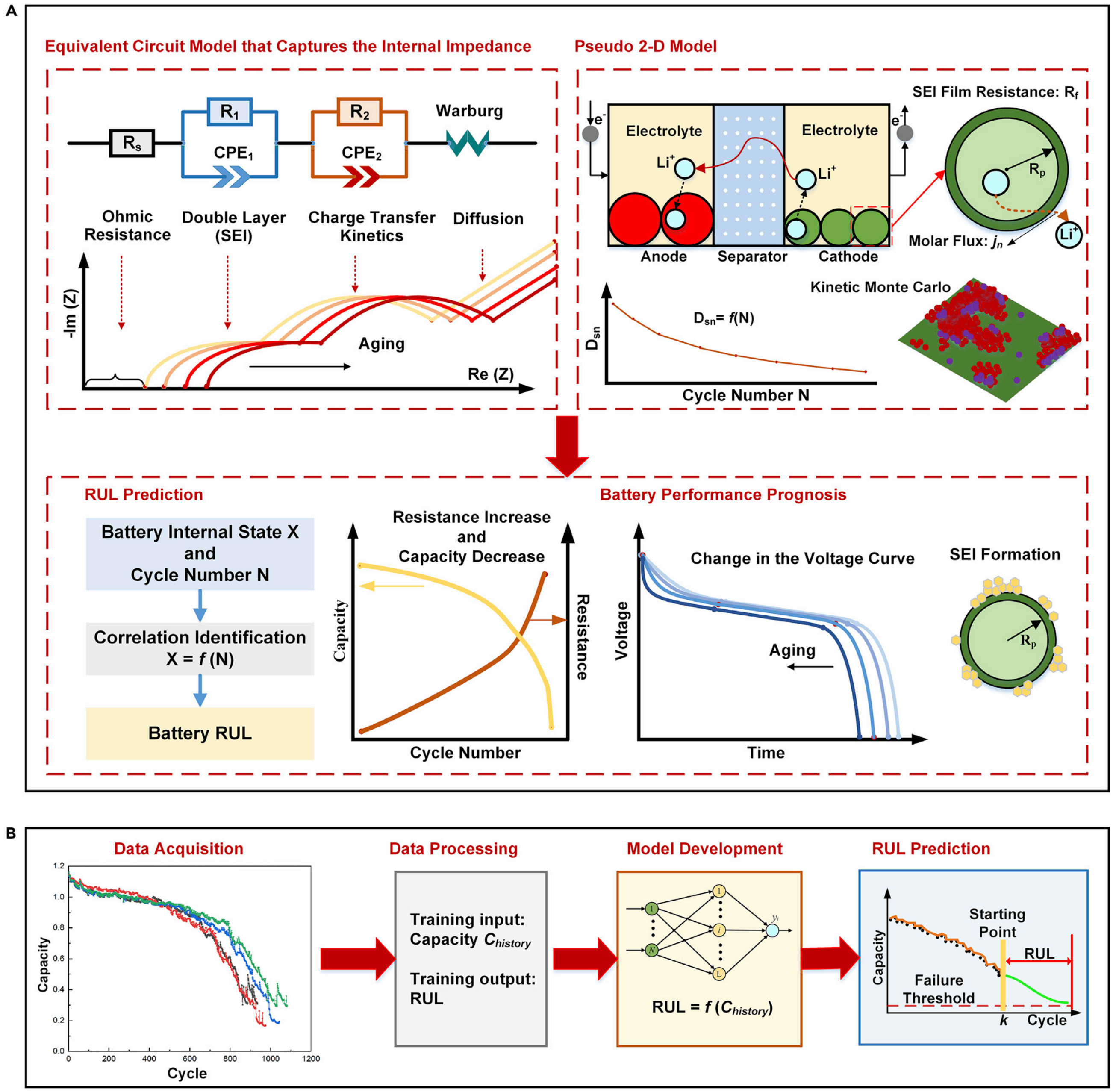

Figure 7. Comparison of Different Prognostic Methods on Battery Performance Prognosis

(A) RUL prognosis methods based on expert knowledge. The ECM consists of an ohmic resistance (represents contact resistance), two ZARC (a resistance and a parallel constant-phase element) elements (represents double layer and charge transfer kinetics), and a Warburg element (represents lithium-ion diffusion within both electrodes). EIS test is used to determine the values of these elements, and their values change as the battery ages (e.g., ohmic resistance increases during battery aging). The P2D model is built using several differential-algebraic equations. For instance, the electrolyte concentration is modeled based on concentrated solution theory and material conservations. The aging behavior is added by using the semi-empirical model or using the mesoscale model, such as kinetic Monte Carlo simulation. The above two methods can not only predict battery RUL but also evaluate battery performance, such as resistance increase and change in the voltage curve.

(B) RUL prognosis methods without expert knowledge. This type of method includes empirical models and data-driven models. They are built purely based on historical data while battery electrochemical mechanisms are neglected. Therefore, they can only be used for RUL prediction.

and mechanistic models can describe the battery aging mechanisms and predict RUL by mapping the internal state to RUL. Therefore, they are not only used for RUL predictions but can also provide the insights into the battery degradation process and evaluate the SOH and peak power capability of the aged battery. Therefore, researchers can not only predict the future performance of the battery (e.g., resistance increment, SEI formation, and change in the voltage curve) without time-consuming aging tests but also optimize battery design to improve expected performance metrics. Empirical and data-driven models fit the nonlinear relationship between historical data and battery RUL without the need to know the electrochemical information. Therefore, these methods provide limited information. More specifically, they only provide the future trends of the training data. For instance, if capacity is used as the training feature, the model prediction output will be the future trajectory of capacity while other battery internal states are unknown. This further confirms our conclusion that the first principle-based RUL prognosis technology is the most promising approach in future research.

Conclusions

Battery degradation is a complex problem, which involves many electrochemical side reactions in anode, electrolyte, and cathode. Operating conditions have a significant impact on the degradation and therefore the battery lifetime. It is of extreme importance to achieve accurate predictions of the remaining battery lifetime under various operating conditions, which is essential for the battery management system to ensure reliable operation and timely maintenance and is also critical for battery second-life applications. This paper provides a timely and comprehensive review of the development of battery lifetime prognostic technologies with a focus on recent advances in model-based, data-driven, and hybrid approaches. The details, advantages, and limitations of these approaches are presented, analyzed, compared, and thoroughly discussed. The training data for the model-based approach are simple (e.g., capacity and internal resistance), whereas the model complexity varies significantly from empirical to mechanistic. The data-driven approaches are popular when there are sufficient data and various direct HIs, and refined HIs can be used to form the training datasets. Generally, the computational complexity of statistical approaches is lower than that of artificial intelligence approaches, whereas the latter approach has a better nonlinear fitting capability. The hybrid approach combines different types of algorithms to introduce new functions such as error correction and data processing to improve prediction accuracy.

Although many studies have investigated the battery RUL prognostics, the RUL prognostics field is still at an early stage of development. The main challenges are related to the usefulness and applicability of RUL prediction approaches for engineering applications. Compared with batteries used in portable electronic devices and independent energy storage devices (e.g., photovoltaic power generation and wind power systems), batteries in the electrified transportation systems and smart grids are subject to dynamic working conditions with strong electromagnetic noise. Therefore, the ability to identify multiple operating modes and adjust algorithms accordingly, as well as robustness against noise, is of vital importance for the battery-powered systems. Furthermore, considering the onboard computing power of the battery management system, achieving a balance between model complexity and accuracy in model-based approaches, decreasing data demands and computational loads in data-driven approaches, and developing more effective frameworks for hybrid approaches are critical for real-time application. Developing the battery RUL prognosis for large-scale lithium-ion battery packs is also a challenge due to the inconsistencies in the battery pack. The inconsistencies among cells make it difficult to develop accurate models for the degradation of the battery pack.

More importantly, in order to ensure that the prognostic methods have good accuracy throughout the entire life cycle under different working conditions, first principle-based prognosis methods involving multi-scale and multi-physics modeling are needed. By incorporating electrochemical, thermal, and mechanical information into RUL prognosis, the first principle-based prognosis model is able to predict battery degradation accurately under different working conditions. However, it is difficult to build due to the high nonlinearity nature of the battery system. The computational complexity of the first principle-based models also hinders their actual applications. Developing the first principle-based RUL prognosis with low computational complexity is one major challenge in future studies.

The rapid development of intelligent transportation systems and connected vehicles provides opportunities to introduce cloud computing and blockchain technology for RUL prediction, which can increase computational efficiency and enrich training data. Finally, to facilitate the utilization of retired lithium-ion batteries, the RUL algorithm available for second-life use needs further study. Second-life use is significant for environmental protection. Otherwise, the remaining resources (e.g., 80% remaining capacity) of a large number of batteries will continue to be wasted.

The battery RUL prognostics field will receive increasing attention as batteries are deployed at larger scale and for more applications than in the past. This paper provides a comprehensive understanding of the development of the battery RUL prediction, which helps the development of RUL algorithms, empowering the safe and reliable implementation of lithium-ion batteries as reliable and efficient energy storage devices. It is also worth mentioning that although this paper mainly focuses on the RUL prognosis of lithium-ion batteries, the methods described in previous sections are applicable to most nonlinear time-variant systems and can therefore be used for other energy storage devices, such as other types of batteries or capacitors.

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AUTHOR CONTRIBUTIONS

X.H. conceived the idea. X.H., L.X., and X.L. conducted the literature review and wrote the manuscript. X.H., L.X., and X.L. discussed and revised the manuscript. M.P. reviewed and edited the manuscript.

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